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Information-Theoretic Causal Bounds under Unmeasured Confounding

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Abstract

We develop an information-theoretic framework for partial identification of causal effects under unmeasured confounding. Existing partial identification approaches suffer from one or more of the following limitations: restricting outcomes to be bounded or discrete, requiring auxiliary inputs (instruments, proxies, or user-specified sensitivity parameters), necessitating full structural modeling, or neglecting effect heterogeneity. We universally address these limitations through a novel information-theoretic divergence bound. Our key insight is that the f-divergences of the observational distribution $P(Y \mid A = a, X = x)$ and the interventional distribution $P(Y \mid \mathrm{do}(A = a), X = x)$ are upper bounded by a function of the propensity score $\Pr(A = a|X = x)$. We translate these upper bounds of f-divergences into sharp lower and upper bounds of conditional causal effects without requiring boundedness assumptions, auxiliary variables, or full data-generating process specification. We develop a semiparametric estimator for the proposed causal bounds that attains fast convergence rates even when nuisance components converge slowly. Simulation studies and real data applications demonstrate the practical utility of our bounds.

Keywords: partial identification, f-divergence, causal bounds, unmeasured confounding

1. Introduction

Causal effect identification aims to represent interventional quantities, such as $\Pr(Y = y \mid \mathrm{do}(A = a), X = x)$ (where A is a treatment, X a covariate and Y an outcome), as functionals of the observational distribution $P(X, A, Y)$. This can be done under strong assumptions (e.g., no unmeasured confounders). When unmeasured confounders U presents as in Fig. 1(*a*), causal effects are in general not point identified from observational data. When point identification fails, *partial identification* can still deliver informative conclusions by constructing sharp lower and upper bounds that provably contain the true causal quantity. However, most existing methods (ref. Sec. 1.1) suffer from one or more of the following limitations:

(Lim-1) Bounded outcomes: Restricting the outcome to be bounded (e.g., $Y \in [0, 1]$) or discrete.

(Lim-2) Auxiliary inputs: Requiring additional auxiliary inputs, including instruments, proxies of unmeasured confounders, or sensitivity parameters quantifying confounding strength.

(Lim-3) Full SCM modeling: Necessitating modeling the entire structural causal models (SCM) (Pearl, 2000), which is computationally intensive and prone to error propagation.

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(Lim-4) Neglecting heterogeneity: Focusing on population-level averages, neglecting covariate-conditional treatment effects.

To universally address these limitations, we develop information-theoretic bounds for conditional causal effects $\mathbb{E}[Y \mid \mathrm{do}(A = a), X = x]$ (and their marginal case $\mathbb{E}[Y \mid \mathrm{do}(A = a)]$) that accommodate unbounded continuous outcomes without requiring full SCM modeling or auxiliary inputs. The key idea is to derive data-driven upper bounds of statistical divergences between the interventional law $Q_{a,x} \triangleq P(Y \mid \mathrm{do}(A = a), X = x)$ and the observational law $P_{a,x} \triangleq P(Y \mid A = a, X = x)$, and then translate these divergence bounds into sharp bounds on causal effects. Specifically, we make three main contributions.

(i) We derive upper bounds on a broad class of f-divergences (Csiszár, 1967) between $P_{a,x}$ and $Q_{a,x}$ that depend only on the propensity score $e_a(x) \triangleq \Pr(A = a \mid X = x)$.

(ii) Using these divergence bounds, we obtain lower and upper bounds on arbitrary expectations of the form $\theta(a, x) \triangleq \mathbb{E}_{Q_{a,x}}[\varphi(Y)]$ for user-specified measurable functionals φ (e.g., CDF values, moments, or loss-based risks) without imposing boundedness of outcomes.

(iii) We develop sample-efficient estimators for the derived causal bounds that are robust to perturbations of nuisance components.

Taken together, these results yield a unified, information-theoretic approach to partial identification of interventional distributions under unmeasured confounding.

1.1. Related Work

We organize existing work on partial identification based on which of the limitations (Lim-(1,2,3,4)) they retain or address. Cited works are not mutually exclusive.

Bounded/discrete outcomes (Lim-1). Early work imposed restrictions requiring outcomes to be bounded or discrete. For example, Manski (1990) derived nonparametric bounds using the extreme values outcomes can attain. Linear-programming (LP)–based approaches (e.g., Balke and Pearl (1994), Tian and Pearl (2000)) yield sharp bounds with discrete variables. Sachs et al. (2023) and Shridharan and Iyengar (2023) have extended these LP-based bounds to general graphical settings but remain restricted to discrete outcomes. Zhang and Bareinboim (2021) extended these LP ideas to continuous outcomes, but still rely on bounded-support assumptions (e.g., $Y \in [0, 1]$). These methods avoid auxiliary inputs (addressing Lim-2) but fail to accommodate unbounded outcomes (Lim-1) or provide conditional effect bounds (Lim-4).

Auxiliary inputs (Lim-2). Another line of work leverages auxiliary inputs. While auxiliary-variable methods can yield sharp bounds, most methods still assume bounded outcomes (Lim-1); and valid auxiliary inputs are often not available in practice or not identifiable from data.

- *Instrumental variables.* Balke and Pearl (1997) provide tight nonparametric bounds on average treatment effects by leveraging instrumental variables, assuming bounded binary outcomes. Kitagawa (2021) extends this framework to continuous outcomes while maintaining bounded support assumptions (see Swanson et al. (2018) for a comprehensive survey). Recently, Levis et al. (2025)

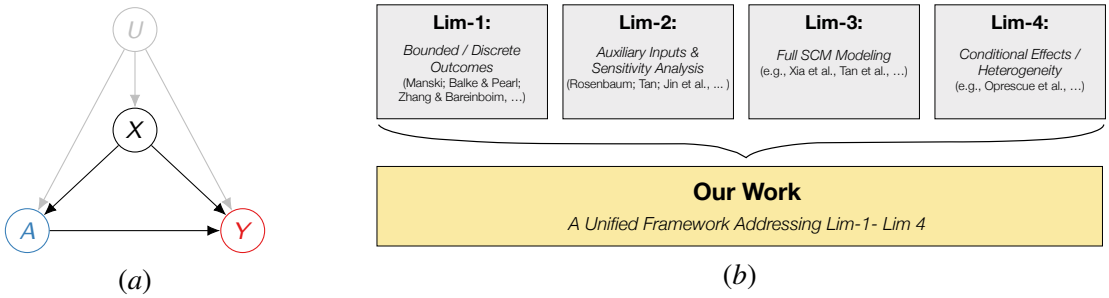

Figure 1: **(a)** Causal diagram with unmeasured confounding. **(b)** Positioning of our method.

develop covariate-assisted IV bounds to target conditional treatment effects (addressing Lim-4), but also under bounded outcome assumptions (Lim-1).

- *Additional assumptions or variables.* Ghassami et al. (2023) leverage proxy variables of hidden confounders to provide bounds on average effects, again requiring bounded outcomes (Lim-1). Lee (2009) and Semenova (2025) avoid bounded outcomes for sharp bounds on the average effect, but rely on structural assumptions about the selection mechanism.

- *Sensitivity analysis.* Sensitivity analysis introduces user-specified parameters to quantify confounding strength (e.g., Rosenbaum (1987), Tan (2006), Yadlowsky et al. (2022), Jin et al. (2022), Dorn and Guo (2023), Oprescu et al. (2023)). Unlike IV and proxy methods, modern sensitivity approaches can accommodate unbounded outcomes (addressing Lim-1). Among these, Jin et al. (2022) are most closely related to our approach, as they use an f-divergence-based sensitivity model to constrain divergences between observational and interventional distributions. Oprescu et al. (2023) extend sensitivity analysis to bound conditional effects (addressing Lim-4). However, all sensitivity methods require external sensitivity parameters (Lim-2) that are not identifiable from observational data alone.

Full SCM-modeling approaches (Lim-3). Another approach leverages machine-learning methods to learn entire SCMs consistent with observational data (e.g., Hu et al. (2021), Balazadeh Meresht et al. (2022), Padh et al. (2023), Xia et al. (2022), Tan et al. (2024)). These approaches find the SCMs that maximize/minimize the target causal effect subject to observations, using flexible neural architectures to model structural functions. In principle, such methods can accommodate unbounded outcomes and target conditional effects (addressing Lim-(1,4)). However, they require estimating the entire SCM (Lim-3), which is computationally intensive and sensitive to misspecification in high-dimensional structural components.

Our novelty. Existing methods each resolve some limitations; however, no existing approach achieves all four limitations (Lim-1–Lim-4) universally. In contrast, our work simultaneously addresses all four limitations by developing bounds that (Lim-1) accommodate unbounded continuous outcomes without support restrictions; (Lim-2) require no auxiliary variables or sensitivity parameters; (Lim-3) avoid full SCM modeling; and (Lim-4) provide bounds for conditional effects $\mathbb{E}[Y \mid \mathrm{do}(A = a), X = x]$ beyond the population-level average. We compare our work with representative existing methods in Fig. 1(*b*).

2. Problem Setup & Background

Let $A \in \{0, 1\}$ be a treatment, $X \in \mathbb{R}^k$ a covariate vector, and $Y \in \mathbb{R}^d$ an outcome. The structural causal model (Pearl, 2000) for (X, A, Y) is

$$U \leftarrow f_U(\epsilon_U), \quad X \leftarrow f_X(U, \epsilon_X), \quad A \leftarrow f_A(X, U, \epsilon_A), \quad Y \leftarrow f_Y(X, A, U, \epsilon_Y), \tag{1}$$

where f_U, f_X, f_A, f_Y are unknown structural functions, $\epsilon \triangleq (\epsilon_U, \epsilon_X, \epsilon_A, \epsilon_Y)$ is a set of mutually independent exogenous variables for the endogeneous variables (U, A, X, Y) with a joint probability law $P(\epsilon)$, and U is a *hidden* common cause of (X, A, Y). The causal diagram induced by this SCM is shown in Fig. 1(*a*). The operation $\mathrm{do}(A = a)$ denotes an intervention that fixes $A = a$ by replacing the structural assignment f_A with the constant assignment $A \leftarrow a$.

For any $a \in \mathcal{A}$ and $x \in \mathcal{X}$, where $\mathcal{A}$ and $\mathcal{X}$ denote the support of A and X, respectively, we define

$$P_{a,x} \triangleq P(Y \mid A = a, X = x), \quad Q_{a,x} \triangleq P(Y \mid \mathrm{do}(A = a), X = x), \tag{2}$$

as the *observational/interventional conditional law* of Y given $A = a$ and $X = x$.

f-Divergence. To measure the discrepancy between $P_{a,x}$ and $Q_{a,x}$, we use the f-divergence:

Definition 1 (f-Divergence (Rényi, 1961; Ali and Silvey, 1966; Csiszár, 1967)) Let P and Q be probability measures on $(\mathcal{Y}, \mathcal{F})$ such that $P \ll Q$. For a convex function $f : [0, \infty) \to \mathbb{R}$ with $f(1) = 0$, the f-divergence of P from Q is

$$D_f(P\|Q) \triangleq \int_{\mathcal{Y}} f\left(\frac{dP}{dQ}\right) dQ = \int_{\mathcal{Y}} f\Big(\frac{p(y)}{q(y)}\Big) q(y) dy. \tag{3}$$

This family includes many familiar quantities, such as Kullback–Leibler (KL) divergence (with $f(t) = t \log t$), Hellinger distance (with $f(t) = \frac{1}{2}(\sqrt{t}-1)^2$), χ^2-divergence (with $f(t) = \frac{1}{2}(t-1)^2$), total variation (TV) distance (with $f(t) = |t - 1|$), and Jensen–Shannon divergence (with $f(t) = t \log t - (t + 1) \log((t + 1)/2) + \log 2$).

Problem Statement. Under the assumed DGP in Fig. 1(*a*), the interventional law $Q_{a,x}$ and a target causal effect in the form of $\mathbb{E}_{Q_{a,x}}[\varphi(Y)]$ (where φ is a user-specified and potentially continuous and unbounded function) are generally not identifiable from the observational law $P_{a,x}$ due to unmeasured confounding. To address, (1) we derive the upper limit of the f-divergence of the observational law $P_{a,x}$ from the interventional law $Q_{a,x}$; i.e., $D_f(P_{a,x}\|Q_{a,x})$; (2) we translate the upper limit of the f-divergence into the causal bounds. Throughout the paper, we assume the following:

Assumption 1 For all $a, x \in \mathcal{A} \times \mathcal{X}$,

1. **Positivity**: $e_a(x) \triangleq \Pr(A = a \mid X = x) \in [c, 1 - c]$ for some constant $0 < c < 1/2$.
2. **Mutual absolute continuity**: $P_{a,x} \ll Q_{a,x}$ and $Q_{a,x} \ll P_{a,x}$.
3. **Finite value at zero**: For the generator function f in the f-divergence, $f(0) < \infty$.

3. Divergence Bounds between Observational and Interventional Distributions

In this section, we derive a data-driven upper bound on the f-divergence between observational and interventional distributions. Our main result is the following:

Theorem 1 (f-Divergence Bound) For any $a \in \mathcal{A}$ and $x \in \mathcal{X}$ such that $P(a \mid x) > 0$,

$$D_f(P_{a,x} \| Q_{a,x}) \leq B_f(e_a(x)), \tag{4}$$

where

$$B_f(e_a(x)) \triangleq e_a(x) f\left(\frac{1}{e_a(x)}\right) + (1 - e_a(x)) f(0) \tag{5}$$

Thm. 1 states that the f-divergence between the observational law $P_{a,x}$ and the interventional law $Q_{a,x}$ is upper bounded by a function of the propensity score, $B_f(e_a(x))$, which is directly computable from observational data without knowledge of the unmeasured confounder. An important property is that $B_f(e_a(x)) \to 0$ as $e_a(x) \to 1$, since f is continuous (by convexity) and satisfies $f(1) = 0$. In other words, higher propensity scores correspond to tighter divergence bounds.

We specialize Thm. 1 to standard divergences:

Corollary T1.1 For any $a \in \mathcal{A}$ and $x \in \mathcal{X}$ such that $P(a \mid x) > 0$,

- **KL**: $f(t) \triangleq t \log t$ (with $f(0) = 0$),

$$D_{\mathrm{KL}}(P_{a,x} \| Q_{a,x}) \leq -\log e_a(x). \tag{6}$$

- **Hellinger**: $f(t) \triangleq \frac{1}{2}(\sqrt{t} - 1)^2$ (with $f(0) = 1/2$),

$$D_{\mathrm{H}}(P_{a,x} \| Q_{a,x}) \leq \min\left(1 - \sqrt{e_a(x)}, -\tfrac{1}{2} \log e_a(x)\right). \tag{7}$$

- **χ^2-divergence**: $f(t) \triangleq \frac{1}{2}(t - 1)^2$ (with $f(0) = 1/2$),

$$D_{\chi^2}(P_{a,x} \| Q_{a,x}) \leq \frac{1 - e_a(x)}{2 e_a(x)}. \tag{8}$$

- **Total variation**: $f(t) \triangleq |t - 1|$ (with $f(0) = 1$),

$$D_{\mathrm{TV}}(P_{a,x} \| Q_{a,x}) \leq \min\left\{1 - e_a(x), \sqrt{-\tfrac{1}{2} \log e_a(x)}\right\}. \tag{9}$$

- **Jensen-Shanon**: $f(t) \triangleq f_{\mathrm{JS}}(t) \triangleq t \log t - (t+1) \log\left(\frac{t+1}{2}\right) + \log 2$ with $f_{\mathrm{JS}}(0) = \log 2$,

$$D_{\mathrm{JS}}(P_{a,x} \| Q_{a,x}) \leq B_{f_{\mathrm{JS}}}(e_a(x)). \tag{10}$$

Beyond the hard intervention $A = a$, bounds extend to stochastic policies as follows:

Corollary T1.2 For any stochastic policy $\pi(a \mid x)$,

$$D_f(P_\pi \| Q_\pi) \triangleq \mathbb{E}_X \left[\sum_{a \in \mathcal{A}} \pi(a \mid X) D_f(P_{a,X} \| Q_{a,X}) \right] \leq \mathbb{E}_X \left[\sum_{a \in \mathcal{A}} \pi(a \mid X) B_f(e_a(X)) \right].$$

Choosing $\pi(a \mid x) = e_a(x)$ yields the global divergence bound:

$$D_f(P_{A,X} \,\|\, Q_{A,X}) = \mathbb{E}_X \left[\sum_{a \in \mathcal{A}} e_a(X)\, D_f(P_{a,X} \,\|\, Q_{a,X}) \right] \;\leq\; \mathbb{E}_X \left[\sum_{a \in \mathcal{A}} e_a(X)\, B_f(e_a(X)) \right].$$

Beyond the f-divergence, we derive bounds on the mean-diff., maximum mean discrepancy (MMD; Gretton et al. 2012), and the integral probability metric (IPM; Müller 1997). See Sec. B for their formal definitions.

Corollary T1.3 (Mean-Diff., MMD, and IPM Bounds) Let $\Phi \triangleq \{\varphi : \mathcal{Y} \mapsto [0,1]\}$ be a class of measurable functions. Let $D_{\mathrm{MMD},\mathtt{k}}(P \| Q)$ be the MMD associated with an RKHS with a kernel $\mathtt{k}$ such that $\mathtt{k}(\cdot,\cdot) < K$. Let $D_{\mathrm{IPM},\mathcal{F}}(P \| Q)$ be the IPM over a function class $\mathcal{F} \triangleq \{f : \|f\|_\infty < C\}$. Then,

$$\textbf{(mean diff.)} \quad \sup_{\varphi \in \Phi} \left| \mathbb{E}_{P_{a,x}}[\varphi(Y)] - \mathbb{E}_{Q_{a,x}}[\varphi(Y)] \right| \leq \min\left\{1 - e_a(x),\, \sqrt{-\tfrac{1}{2}\log e_a(x)}\right\},$$

$$\textbf{(MMD)} \quad D_{\mathrm{MMD},\mathtt{k}}(P_{a,x} \,\|\, Q_{a,x}) \leq \sqrt{K}\, \min\left\{1 - e_a(x),\, \sqrt{-\tfrac{1}{2}\log e_a(x)}\right\},$$

$$\textbf{(IPM)} \quad D_{\mathrm{IPM},\mathcal{F}_C}(P_{a,x} \,\|\, Q_{a,x}) \leq C\, \min\left\{1 - e_a(x),\, \sqrt{-\tfrac{1}{2}\log e_a(x)}\right\}$$

Additionally, we derive closed-form expressions for the f-divergence for several exponential-family models (e.g., Bernoulli, Gaussian, Poisson, exponential) to illustrate the mechanism of Thm. 1 in Sec. A.1 of the Appendix.

4. From Divergence Bounds to Causal Effect Bounds

In this section, we derive sharp bounds on the target causal effect of the form $\theta(a,x) \triangleq \mathbb{E}_{Q_{a,x}}[\varphi(Y)]$, where $\varphi(Y)$ is an arbitrary measurable function with finite first and second moments. This framework encompasses diverse causal quantities: setting $\varphi(Y) \triangleq \mathbf{1}(Y \leq t)$ yields the cumulative distribution function $Q_{a,x}(Y \leq t)$, while choosing $\varphi(Y) \triangleq \ell(Y;\theta)$ (a loss function for θ) yields the risk function over $Q_{a,x}$. Crucially, we impose no restrictions requiring φ to be discrete or bounded.

To derive the bounds, we define the f-divergence-based *ambiguity set*, which is a collection of distributions over Y within the $B_f(e_a(x))$ radius around the observational law $P_{a,x}$:

$$\textbf{(Ambiguity set)} \quad \mathcal{Q}_f(a,x;P_{a,x}) \triangleq \left\{ Q_{a,x} \in \mathcal{P}_{a,x}(\mathcal{Y}) \;:\; \begin{array}{l} D_f(P_{a,x} \,\|\, Q_{a,x}) \leq B_f(e_a(x)), \\ P_{a,x} \ll Q_{a,x} \end{array} \right\}, \quad (11)$$

where $\mathcal{P}_{a,x}(Y)$ is a collection of probability laws given $A = a$ and $X = x$. The target causal effect $\mathbb{E}_{Q_{a,x}}[\varphi(Y)]$ is bounded by expectations over the extremal distributions in this ambiguity set:

$$\textbf{(Bounds)} \qquad \underbrace{\theta_{\text{lo}}(a,x)}_{\inf_{Q\in\mathcal{Q}_f(a,x)} \mathbb{E}_Q[\varphi(Y)]} \leq \underbrace{\theta(a,x)}_{\mathbb{E}_{Q_{a,x}}[\varphi(Y)]} \leq \underbrace{\theta_{\text{up}}(a,x)}_{\sup_{Q\in\mathcal{Q}_f(a,x)} \mathbb{E}_Q[\varphi(Y)]} \tag{12}$$

Without loss of generality, we focus on the upper bound $\theta_{\text{up}}(a,x)$, the lower bound $\theta_{\text{lo}}(a,x)$ is equivalent to negation of the upper bound with negation of φ (i.e., $-\theta_{\text{up}}(a,x;-\varphi)$; see Prop. 4).

Computing $\theta_{\text{up}}(a,x)$ directly from Eq. (12) is infeasible, as it requires optimizing over all measures in $\mathcal{Q}_f(a,x)$. For computational feasibility, we recast the problem into tractable formulations:

Theorem 2 (Primal and Dual) Let $s(Y) \triangleq \frac{dQ_{a,x}}{dP_{a,x}}(Y)$, $g_s(Y) \triangleq s(Y)\cdot f(1/s(Y))$, and $\eta_f(a,x) \triangleq B_f(e_a(x))$. Then, the primal and dual representation of $\theta_{\text{up}}(a,x)$ are

$$\theta_{\text{up}}(a,x) = \sup_{s>0}\Big\{\mathbb{E}_{P_{a,x}}[s(Y)\varphi(Y)] \;\text{ s.t. } \mathbb{E}_{P_{a,x}}[s(Y)] = 1,\; \mathbb{E}_{P_{a,x}}[g_s(Y)] \leq \eta_f(a,x)\Big\} \tag{13}$$

$$= \inf_{\lambda>0,u\in\mathbb{R}} \{\lambda\eta_f(a,x) + u + \lambda\mathbb{E}_{P_{a,x}}\Big[g^*\big(\tfrac{\varphi(Y)-u}{\lambda}\big)\Big]\}, \tag{14}$$

where $g^*(t) \triangleq \sup_{s>0}\{s\,t - g(s)\}$ is the convex conjugate (also known as the Legendre–Fenchel conjugate or c-transform) of g.

The following proposition provides a general recipe for computing the convex conjugate g^*:

Proposition 1 (Convex Conjugate g^*) Let $f : (0,\infty) \to (-\infty,+\infty]$ be a proper, convex, lower–semicontinuous function. Define $g(s) \triangleq s\,f(1/s), \quad s > 0$, and $r \triangleq 1/s > 0$. Let $\partial f(r)$ denote the subdifferential of f at r, i.e. $\partial f(r) \triangleq \{a : \; f(u) \geq f(r) + a(u-r) \;\; \forall u \geq 0\}$. Define the trajectory map as $T(r) \triangleq \{f(r) - r\,\partial f(r)\}$ and $\mathcal{T} \triangleq \bigcup_{r>0} T(r)$. Let $g^*(t) \triangleq \sup_{s>0}\{st - g(s)\}$. Then the following hold:

1. Suppose f is continuously differentiable, and the trajectory map $t(r) \triangleq f(r) - rf'(r)$ is invertible with inverse $r(\cdot)$. Then, for every t in the range of $t(\cdot)$,

$$g^*(t) = -f'(r(t)). \tag{15}$$

2. For general f, we have

$$g^*(t) = \begin{cases} -a & \text{if } t \in \mathcal{T} \\ \max\Big\{\limsup_{r\downarrow 0} \frac{t-f(r)}{r}, \limsup_{r\uparrow\infty} \frac{t-f(r)}{r}\Big\} & \text{otherwise.} \end{cases} \tag{16}$$

We specialize Proposition 1 to standard f-divergences:

Corollary P1 Let $g(s) \triangleq sf(1/s)$ for $s > 0$. Then,

- **KL**: $g_{\mathrm{KL}}(s) = -\log s$, $g^*_{\mathrm{KL}}(t) = -1 - \log(-t)$ if $t < 0$; $g^*_{\mathrm{KL}}(t) = \infty$ if $t \geq 0$.
- **Hellinger**: $g_{\mathrm{H}}(s) = \frac{1}{2}(1 - 2\sqrt{s} + s)$. $g^*_{\mathrm{H}}(t) = \frac{t}{1-2t}$ if $t < 1/2$; $g^*_{\mathrm{H}}(t) = \infty$ if $t \geq 1/2$.
- **Chi-square**: $g_{\chi^2}(s) = \frac{(1-s)^2}{2s}$. $g^*_{\chi^2}(t) = -(1+\sqrt{1-2t})$ if $t \leq 1/2$; $g^*_{\chi^2}(t) = \infty$ if $t > 1/2$.
- **TV**: $g_{\mathrm{TV}}(s) = |1-s|$. $g^*_{\mathrm{TV}}(t) = -\mathbf{1}(t \leq -1) + t\mathbf{1}(t \in (-1, 1])$ if $t \leq 1$; $g^*_{\mathrm{TV}}(t) = \infty$ if $t > 1$.
- **Jensen-Shanon**: $g_{\mathrm{JS}}(s) = sf_{\mathrm{JS}}(1/s)$; $g^*_{\mathrm{JS}}(t) = \log\left(\frac{2}{4-e^t}\right)$ if $t < \log 4$; $g^*_{\mathrm{JS}}(t) = \infty$ if $t \geq \log 4$.

5. Semiparametric Estimation and Inference

A computational challenge in solving the dual problem in Eq. (14) is that each optimization iteration requires evaluating $g^*(\cdot)$ and computing the conditional expectation $\mathbb{E}_{P_{a,x}}[g^*] = \mathbb{E}[g^* \mid A = a, X = x]$, typically through regression, rendering direct optimization computationally infeasible. To circumvent this challenge, we consider functional parameters $\lambda(a, x) > 0$ and $u(a, x) \in \mathbb{R}$ instead of optimizing λ, u separately for each (a, x). To enforce positivity, we parameterize $\lambda(a, x) = \exp(h(a, x))$ for $h(a, x) \in \mathbb{R}$. The dual problem then becomes:

Proposition 2 Let $\eta_f(a, x) \triangleq B_f(e_a(x))$. Then,

$$\theta_{\mathrm{up}}(a, x) = \inf_{\substack{h(a,x)\in\mathbb{R} \\ u(a,x)\in\mathbb{R}}} \mathbb{E}_{P_{a,x}}\Big[\exp(h(A, X))\{\eta_f(A, X) + g^*\big(\tfrac{\varphi(Y)-u(A,X)}{\exp(h(A,X))}\big)\} + u(A, X)\Big]. \tag{17}$$

To operationalize this optimization, we define a loss function and corresponding risk function for the functional parameters h and u:

Definition 2 (Risk Function for Causal Bound) Let $V = (X, A, Y)$. Let $h_\beta, u_\gamma : \mathcal{A} \times \mathcal{X} \mapsto \mathbb{R}$ be maps parametrized by $\beta \in \mathbb{R}^{p_1}$ and $\gamma \in \mathbb{R}^{p_2}$. The risk function for causal bounds is

$$\mathcal{R}(\beta, \gamma; e) \triangleq \mathbb{E}_P[\ell(V; (\beta, \gamma), e)], \tag{18}$$

where $e \triangleq e_A(X)$ and $\eta_f \triangleq \eta_f(A, X) \triangleq B_f(e_A(X))$, and

$$\ell(V; (\beta, \gamma), e) \triangleq \exp(h_\beta(A, X))\Big\{\eta_f(A, X) + g^*\Big(\tfrac{\varphi(Y)-u_\gamma(A,X)}{\exp(h_\beta(A,X)}\Big)\Big\} + u_\gamma(A, X). \tag{19}$$

The following proposition establishes that minimizing this risk function solves the dual problem:

Proposition 3 (Justification of Risk Function) Define, for each (a, x), the following loss

$$\ell(h, u; y, a, x) \triangleq \exp(h(a, x))\Big\{\eta_f(a, x) + g^*\big(\tfrac{\varphi(y)-u(a,x)}{\exp(h(a,x))}\big)\Big\} + u(a, x). \tag{20}$$

Let $\mathcal{R}(h, u) \triangleq \mathbb{E}_P[\ell(h, u; Y, A, X)]$ be a risk function. Assume $\mathcal{R}(h, u) < \infty$ for all $h, u \in \mathcal{F}$, where $\mathcal{F}$ is a function class rich enough that for any $(h_1, u_1), (h_2, u_2) \in \mathcal{F}$ and $\forall B \subset \mathcal{A} \times \mathcal{X}$, $(h', u') \triangleq (h_1, u_1)\mathbf{1}_B + (h_2, u_2)\mathbf{1}_{B^c}$ also lies in $\mathcal{F}$. Then, for any fixed $(h^\star, u^\star) \in \mathcal{F}$, the followings are equivalent:

1. $(h^\star, u^\star)$ minimizes $\mathcal{R}$ over $\mathcal{F}$.
2. $(h^\star, u^\star)$ minimizes $\mathbb{E}_{P_{a,x}}[\ell(h, u; Y, a, x)]$ for $P_{A,X}$-almost every (a, x).

Proposition 3 shows that solving Eq. (17) is equivalent to finding the minimizer $(h^\star, u^\star)$ of the risk function in Def. 2, a substantially more tractable optimization problem.

The risk function in Eq. (18) depends on the propensity score e, which must be estimated from data. To ensure robustness to estimation errors in e, we construct a debiased risk function that achieves first-order insensitivity (Neyman-orthogonality) with respect to perturbations in the propensity score:

Definition 3 (Debiased Risk) Let $\eta_f'(A, X)$ be the first-order derivative of $\eta_f(A, X)$ w.r.t. e. The debiased risk function is

$$\mathcal{R}^{\mathrm{db}}(\beta, \gamma; e) \triangleq \mathbb{E}[\ell^{\mathrm{db}}(V; (\beta, \gamma), e)], \tag{21}$$

where

$$\ell^{\mathrm{db}}(V; (\beta, \gamma), e) \triangleq \underbrace{\exp(h_\beta(A, X))\Big\{\eta_f(A, X) + g^*\big(\tfrac{\varphi(Y)-u_\gamma(A,X)}{\exp(h_\beta(A,X))}\big)\Big\} + u_\gamma(A, X)}_{\text{Eq. (19)}} \tag{22}$$

$$+ \sum_{a\in\mathcal{A}} e_a(X)\exp(h_\beta(a, X))\eta_f'(a, X)\left(\mathbf{1}(A = a) - e_a(X)\right) \tag{23}$$

This debiased risk function satisfies Neyman-orthogonality (Chernozhukov et al., 2018) with respect to the propensity score, ensuring that the risk is invariant to first-order perturbations in e:

Lemma 1 (Orthogonality) For any direction functions $\{s_a(\cdot)\}_{a\in\mathcal{A}}$ and any perturbation path $e_{t,a} \triangleq e_a + ts_a$ with sufficiently small $|t|$, $\frac{\partial}{\partial t}\mathcal{R}^{\mathrm{db}}(\beta, \gamma; e_t)\big|_{t=0} = 0$ for all (β, γ).

We now present our estimation procedure based on cross-fitting:

Definition 4 (Debiased Causal Bound Estimators) Fix a functional φ and an f-divergence. Let ℓ^{db} and $\mathcal{R}^{\mathrm{db}}$ be as in Def. 3. The debiased estimator of the upper causal bound $\theta_u(a, x)$ for any $(a, x) \in \mathcal{A} \times \mathcal{X}$ is constructed as follows:

1. Randomly split the dataset $\mathcal{D}$ (with size n) into K disjoint folds $\mathcal{D}_1, \cdots, \mathcal{D}_K$.

2. For each k fold, learn $\widehat{e}_a^k$ using $\mathcal{D}_{-k} \triangleq \mathcal{D} \setminus \mathcal{D}_k$ for all $a \in \mathcal{A}$.

3. For each fold k, solve $\widehat{\vartheta}_k \triangleq (\hat{\beta}_k, \hat{\gamma}_k) \in \arg\min_{\beta,\gamma} \sum_{i|V_i \in \mathcal{D}_k} \ell^{\mathrm{db}}(V_i; (\beta, \gamma), \widehat{e}^k)$.

4. For each fold k, evaluate

$$\widehat{h}_k(a, x) \triangleq h_{\widehat{\beta}_k}(a, x), \qquad \widehat{u}_k(a, x) \triangleq u_{\widehat{\gamma}_k}(a, x), \tag{24}$$

$$\widehat{\lambda}_k(a, x) \triangleq \exp\{\widehat{h}_k(a, x)\}, \qquad \widehat{\eta}_f^k(a, x) \triangleq B_f(\widehat{e}_a^k(x)). \tag{25}$$

5. For each fold k and each $i \in \mathcal{D}_k$, evaluate $Z_i^k \triangleq g^*\big(\frac{\varphi(Y_i) - \widehat{u}_k(A_i, X_i)}{\widehat{\lambda}_k(A_i, X_i)}\big)$, and learn a regressor $\widehat{m}_k$ by regressing Z_i^k onto (A, X) using $\mathcal{D}_k$.

6. Evaluate $\widehat{\theta}_{\mathrm{up}}^{(k)}(a, x) \triangleq \widehat{\lambda}_k(a, x)\big(\widehat{\eta}_f^k(a, x) + \widehat{m}_k(a, x)\big) + \widehat{u}_k(a, x)$ and return $\widehat{\theta}_{\mathrm{up}}(a, x) \triangleq (1/K) \sum_{k=1}^K \widehat{\theta}_{\mathrm{up}}^{(k)}(a, x)$.

The following theorem establishes the convergence rate of our proposed estimator:

Theorem 3 (Error Analysis) Under Assumption 2 (in Sec. A), fix a fold k. Let e be the true propensity score, and $\vartheta_0 \triangleq (\beta_0, \gamma_0) \in \arg\min_\vartheta \mathcal{R}^{\mathrm{db}}(\vartheta; e)$ for $\vartheta \triangleq (\beta, \gamma)$. Let $\widehat{\vartheta}_k \triangleq (\widehat{\beta}_k, \widehat{\gamma}_k)$ be the minimizer from Step 3 in Def. 4 with $\widehat{e}^k$. Define $r_n \triangleq O_P(\|\widehat{e}^k - e\|_2)$. Then,

$$\|\widehat{\vartheta}_k - \vartheta_0\|_2^2 = O_p(n^{-1/2} + r_n^2). \tag{26}$$

Furthermore, let $Z_\vartheta \triangleq g^*\big(\frac{\varphi(Y) - u_\gamma(A,X)\}}{\exp(h_\beta(A,X))}\big)$ and $m_\vartheta(a, x) \triangleq \mathbb{E}[Z_\vartheta \mid A = a, X = x]$. Define $s_n \triangleq O_P(\|\widehat{m}_k - m_{\widehat{\vartheta}_k}\|_2)$ where $\widehat{m}_k$ is from Step 5 in Def. 4. Let $\widehat{\theta}_{\mathrm{up}}^{(k)}$ be the estimated upper causal bound for the fold k. Under additional Assumption 3 (in Sec. A),

$$\|\widehat{\theta}_{\mathrm{up}}^{(k)} - \theta_{\mathrm{up}}\|_2^2 = O_p(n^{-1/2} + r_n^2 + s_n^2). \tag{27}$$

Thm. 3 demonstrates the sample efficiency of our debiased estimator. Even when the nuisance components (the propensity score and the pseudo-outcome regression) converge slowly (e.g., at rate $n^{-1/4}$), both the dual parameters $\widehat{\vartheta}_k$ and the upper-bound estimator $\widehat{\theta}_{\mathrm{up}}^{(k)}$ achieve the faster rate (e.g., at rate $n^{-1/2}$). Specifically, the sample-efficiency gain is of order $O_P(r_n^2)$ rather than $O_P(r_n)$ that would result from using the non-debiased risk function (Eq. (18)). This improvement stems from the orthogonal construction of the debiased risk, which eliminates first-order sensitivity to propensity score errors (Lemma 1). Consequently, nuisance components can be estimated using flexible machine learning methods while the estimator retains faster convergence rates.

5.1. Ensemble Bound Aggregation

Let $\mathcal{F}$ denote a collection of f-generators (e.g., $\mathcal{F} = \{f_{\mathrm{KL}}, f_{\mathrm{TV}}, f_{\chi^2}, \cdots\}$). We obtain a family of upper bound estimates $\widehat{\boldsymbol{\theta}}_{\mathrm{up}} \triangleq \{\widehat{\theta}_{\mathrm{up},f} : f \in \mathcal{F}\}$ where $\widehat{\theta}_{\mathrm{up},f}$ is an upper-bound estimate for certain f.

Let $\widehat{\boldsymbol{\theta}}_{\text{lo}}$ be defined similarly. Our goal is to aggregate these bounds to construct the tightest interval. Ideally, we would select $\widehat{\theta}^{\star}_{\text{up}} = \min_{f\in\mathcal{F}}\{\widehat{\theta}_{\text{up},f}\}$ if every estimates satisfy $\theta(a,x) \le \widehat{\theta}_{\text{up},f}(a,x)$ for any fixed (a,x). However, due to finite samples and numerical instability in dual optimization, some estimates may violate the bound condition (i.e., $\theta(a,x) \ge \widehat{\theta}_{u,f}(a,x)$ for some $f \in \mathcal{F}$ and (a,x)). To ensure robustness against such violations, we propose the following aggregation strategy.

Definition 5 (k-th order statistics aggregator) Let $\widehat{\boldsymbol{\theta}}_{\text{lo}}, \widehat{\boldsymbol{\theta}}_{\text{up}}$ denote candidate lower and upper bounds, respectively, with $n_f \triangleq |\widehat{\boldsymbol{\theta}}_{\text{lo}}| = |\widehat{\boldsymbol{\theta}}_{\text{up}}|$. For $k \in \{1,\dots,n_f\}$, the *k-th order-statistics aggregator* (k-agg) is defined as the pair $(\widehat{\theta}^k_{\text{lo}}, \widehat{\theta}^k_{\text{up}})$, where $\widehat{\theta}^k_{\text{lo}}$ is the k-th smallest element of $\widehat{\boldsymbol{\theta}}_{\text{lo}}$ and $\widehat{\theta}^k_{\text{up}}$ is the k-th largest element of $\widehat{\boldsymbol{\theta}}_{\text{up}}$.

In practice, the k-agg is implemented by initializing $k = n_f$ (selecting the tightest bounds) and iteratively decrementing $k \leftarrow k-1$ until $\widehat{\theta}^k_{\text{lo}} \le \widehat{\theta}^k_{\text{up}}$.

Lemma 2 (Valid Coverage under Partial Correctness) For a fixed (a,x), suppose at least k lower bound estimates in $\widehat{\boldsymbol{\theta}}_{\text{lo}}$ satisfy $\widehat{\theta}_{\text{lo},f}(a,x) \le \theta(a,x)$, and at least k upper bound estimates in $\widehat{\boldsymbol{\theta}}_{\text{up}}$ satisfy $\widehat{\theta}_{\text{up},f}(a,x) \ge \theta(a,x)$. Let $(\widehat{\theta}^k_{\text{lo}}, \widehat{\theta}^k_{\text{up}})$ be the k'th order-statistics aggregator. Then, $\widehat{\theta}^k_{\text{lo}}(a,x) \le \theta(a,x) \le \widehat{\theta}^k_{\text{up}}(a,x)$.

6. Empirical Validation

In this section, we evaluate our proposed causal bounds through synthetic and semi-synthetic experiments. All experiments use the k-agg (with $k = 5$) aggregator (Def. 5) with incremental decrementing, and nuisance functions are estimated using XGBoost (Chen and Guestrin, 2016).

Synthetic Data. We generate data from the structural causal model depicted in Fig. 1(*a*) with $n = 1000$ observations. Covariates $X \in \mathbb{R}^5$ are drawn from a standard multivariate Gaussian. The unmeasured confounder U and treatment A are generated as $U \sim \mathcal{N}(0,1)$ and $A \sim \text{Bernoulli}(\sigma(X_1+U))$, where $\sigma(\cdot)$ is the sigmoid function. The outcome follows $Y = f_Y(A,X)+\varepsilon$ with $\varepsilon \sim \mathcal{N}(0,1)$ in panel (a), and ε following a t-distribution with 3 degrees of freedom in panel (b). The heavy-tailed noise setting demonstrates that our bounds remain valid without sub-Gaussian assumptions on the outcome distribution, which is a key advantage over methods requiring bounded or light-tailed outcomes. We target the conditional average treatment effect $\theta(A=1,x) \triangleq \mathbb{E}[Y \mid \text{do}(A=1), X=x]$. Fig. 2(a–b) display the estimated bounds across values of X_1. The true causal effect (dashed black) lies within the estimated bounds (blue: lower; green: upper), confirming valid coverage. The bounds remain informative across the covariate range, demonstrating our method's ability to capture effect heterogeneity without bounded outcome assumptions.

Semi-Synthetic Data: IHDP. We use the Infant Health and Development Program (IHDP) dataset (Hill, 2011), a widely-used benchmark for causal inference. The dataset provides 10 simulation realizations, each containing $n = 747$ observations with 25 covariates describing infant and maternal characteristics; we select two representative realizations for our experiments. Instead of using all

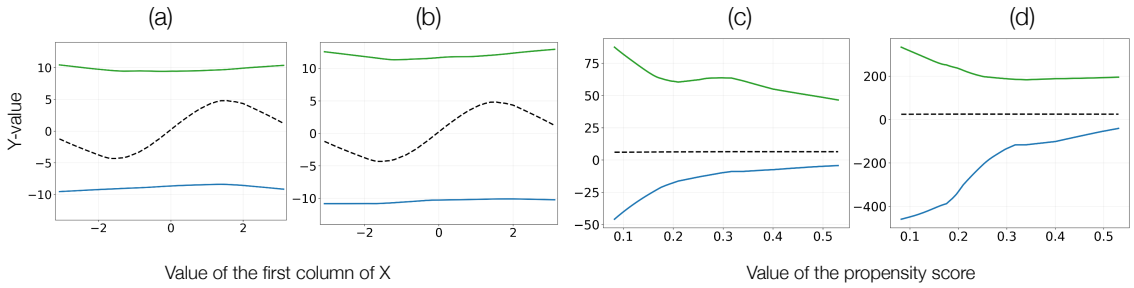

Figure 2: **(a–b)** Synthetic data: estimated upper bound $\hat{\theta}_{\text{up}}(A = 1, x)$ (green) and lower bound $\hat{\theta}_{\text{lo}}(A = 1, x)$ (blue) for the conditional causal effect $\theta(A = 1, x)$, plotted against the first covariate X_1. The true effect is plotted with the dashed black. Panel (a) uses Gaussian noise, and panel (b) uses heavy-tailed t_3-distributed noise. **(c–d)** Semi-synthetic IHDP data (Hill, 2011; Louizos et al., 2017): bounds plotted against the estimated propensity score $\hat{e}_1(x)$. Consistent with Thm. 1, bounds tighten as $e_1(x) \to 1$. Panels (c) and (d) correspond to two different simulation realizations from the IHDP benchmark.

25 covariates, we only use 5 covariates to construct the upper bounds, thereby treating the remaining 20 covariates as unmeasured confounders that affect both treatment assignment and outcomes. Fig. 2(c–d) show the estimated bounds as a function of the propensity score $e_1(x) = \Pr(A = 1 \mid X = x)$. Consistent with Thm. 1, the bounds tighten as the propensity score approaches 1, since the divergence bound $B_f(e_a(x)) \to 0$ as $e_a(x) \to 1$. The true causal effect (dashed black) remains within the bounds across all propensity score values.

Summary. These experiments demonstrate that our information-theoretic bounds (i) provide valid coverage of the true causal effect, (ii) adapt to effect heterogeneity across covariates, (iii) tighten appropriately with increasing propensity scores, and (iv) apply to unbounded continuous outcomes without requiring auxiliary sensitivity parameters, including heavy-tailed distributions that violate sub-Gaussian assumptions commonly required by existing methods.

7. Conclusion and Discussion

We developed an information-theoretic framework for partial identification of causal effects under unmeasured confounding. Our key contribution is deriving data-driven upper bounds on f-divergences between observational and interventional distributions that depend only on the propensity score. These divergence bounds translate into causal effect bounds that simultaneously address four limitations of existing methods: (1) they accommodate unbounded continuous outcomes, (2) require no auxiliary variables or sensitivity parameters, (3) avoid full structural causal model specification, and (4) provide heterogeneous effect bounds conditional on covariates. We developed debiased semiparametric estimators with fast convergence rates even when nuisance components are estimated at slower rates. Empirical results on synthetic and semi-synthetic data confirm valid coverage and demonstrate applicability to heavy-tailed outcome distributions. Future directions include extending the framework to continuous treatments and leveraging tighter divergence bounds through additional structural assumptions.

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Supplement of Information-Theoretic Causal Bounds under Unmeasured Confounding

Appendix A. Omitted Results

A.1. Omitted Results for Sec. 3

A.1.1. Specialization for Exponential Family

In this section, we derive a closed-form f-divergence when $P_{a,x},\ Q_{a,x}$ are within the exponential family (e.g., Bernoulli, Gaussian, Poission, exponential, etc.) to exemplify the mechanism of Thm. 1.

Corollary 5 (Exponential Family) Suppose $P_{a,x}$ and $Q_{a,x}$ are distributions from a common exponential family:

$$P_{a,x}(y) \triangleq \exp\left(\theta_p^\top T(y) - A(\theta_p)\right)h(y), \tag{28}$$

$$Q_{a,x}(y) \triangleq \exp\left(\theta_q^\top T(y) - A(\theta_q)\right)h(y), \tag{29}$$

where θ_p, θ_q are natural parameters, $T(y)$ is the sufficient statistics, $A(\theta)$ is the log-partition function (log normalizer), and $h(y)$ is the base measure density. Define $\Delta \triangleq \theta_p - \theta_q$ and $\Delta_A \triangleq A(\theta_p) - A(\theta_q)$.

$$D_f(P_{a,x} \| Q_{a,x}) = \mathbb{E}_Q\left[f\left(\exp\left(\Delta^\intercal T(Y) - \Delta_A\right)\right)\right]. \tag{30}$$

Bernoulli Distribution Suppose $Y \in \{0, 1\}$ and both $P_{a,x}$ and $Q_{a,x}$ are Bernoulli distributions with success probabilities p and q, respectively. The Bernoulli distribution belongs to the exponential family with sufficient statistic $T(y) = y$ and natural parameter $\theta = \log \frac{p}{1-p}$.

The Radon–Nikodym derivative is given by

$$\frac{dP_{a,x}}{dQ_{a,x}}(y) = \exp\left(y \log \frac{p(1-q)}{q(1-p)} + \log \frac{1-p}{1-q}\right). \tag{31}$$

Consequently, the f-divergence admits the representation

$$D_f(P_{a,x} \| Q_{a,x}) = \mathbb{E}_{Q_{a,x}}\left[f\left(\exp\left(Y \log \frac{p(1-q)}{q(1-p)} + \log \frac{1-p}{1-q}\right)\right)\right], \tag{32}$$

which matches the general form in Theorem 1.

Gaussian Distribution Suppose $Y \in \mathbb{R}^d$ and both $P_{a,x}$ and $Q_{a,x}$ are Gaussian distributions with means μ_p, μ_q and covariance matrices Σ_p and Σ_q, respectively. In this case, the Gaussian distribution forms an exponential family with sufficient statistic $T(y) = (y, yy^\top)$.

The Radon–Nikodym derivative is given by

$$\frac{dP_{a,x}}{dQ_{a,x}}(y) = \frac{|\Sigma_q|^{1/2}}{|\Sigma_p|^{1/2}} \exp\left(-\frac{1}{2}(y-\mu_p)^\top \Sigma_p^{-1}(y-\mu_p) + \frac{1}{2}(y-\mu_q)^\top \Sigma_q^{-1}(y-\mu_q) \right). \tag{33}$$

Accordingly, using (33), the f-divergence can be written as

$$D_f(P_{a,x} \| Q_{a,x}) = \mathbb{E}_{Q_{a,x}}\left[f\left(\frac{dP_{a,x}}{dQ_{a,x}}(Y) \right) \right], \tag{34}$$

Poisson Distribution Suppose $Y \in \{0, 1, 2, \dots\}$ and both $P_{a,x}$ and $Q_{a,x}$ are Poisson distributions with rate parameters λ_p and λ_q, respectively. The Poisson distribution belongs to the exponential family with sufficient statistic $T(y) = y$ and natural parameter $\theta = \log \lambda$.

The Radon–Nikodym derivative takes the form

$$\frac{dP_{a,x}}{dQ_{a,x}}(y) = \exp\left(y \log \frac{\lambda_p}{\lambda_q} - (\lambda_p - \lambda_q) \right). \tag{35}$$

The corresponding f-divergence is therefore

$$D_f(P_{a,x} \| Q_{a,x}) = \mathbb{E}_{Q_{a,x}}\left[f\left(\exp\left(Y \log \frac{\lambda_p}{\lambda_q} - (\lambda_p - \lambda_q) \right) \right) \right]. \tag{36}$$

Exponential Distribution Suppose $Y \geq 0$ and both $P_{a,x}$ and $Q_{a,x}$ follow exponential distributions with rate parameters λ_p and λ_q, respectively. The exponential distribution is an exponential family with sufficient statistic $T(y) = y$ and natural parameter $\theta = -\lambda$.

For $y \geq 0$, the Radon–Nikodym derivative is given by

$$\frac{dP_{a,x}}{dQ_{a,x}}(y) = \frac{\lambda_p}{\lambda_q} \exp\left(-(\lambda_p - \lambda_q) y \right). \tag{37}$$

Accordingly, the f-divergence can be expressed as

$$D_f(P_{a,x} \| Q_{a,x}) = \mathbb{E}_{Q_{a,x}}\left[f\left(\frac{\lambda_p}{\lambda_q} \exp\left(-(\lambda_p - \lambda_q) Y \right) \right) \right]. \tag{38}$$

Across all examples, the Radon–Nikodym derivative reduces to an exponential function of the sufficient statistic. This highlights how the f-divergence in Theorem 1 depends solely on transformations of sufficient statistics within the exponential-family framework.

A.2. Omitted Results for Sec. 4

We show that deriving a lower bound is a subproblem of deriving an upper bound:

Proposition 4 (Lower bound as a subproblem of upper bound) For any arbitrary function f over some space $\mathcal{X}$, the following holds:

$$\inf_{x\in\mathcal{X}} f(x) \;=\; -\sup_{x\in\mathcal{X}} \big(-f(x)\big). \tag{39}$$

Proof [(Prop. 4)] For any $x \in \mathcal{X}$,

$$-f(x) \le -\inf_{x'} f(x'), \;\; \forall x \in \mathcal{X} \implies \inf_{x\in\mathcal{X}} f(x) \le -\sup_{x\in\mathcal{X}}(-f(x)). \tag{40}$$

Also, by the definition of infimum, for any $\varepsilon > 0$, there exists x_ϵ such that

$$f(x_\varepsilon) \le \inf_{x\in\mathcal{X}} f(x) + \varepsilon. \tag{41}$$

Then,

$$-f(x_\varepsilon) \ge -\inf_{x} f(x) - \varepsilon \implies \sup_{x\in\mathcal{X}}(-f(x)) \ge -\inf_{x\in\mathcal{X}} f(x) - \varepsilon. \tag{42}$$

By taking $\epsilon \downarrow 0$, we have $\sup_{x\in\mathcal{X}}(-f(x)) \ge -\inf_{x\in\mathcal{X}} f(x)$. The proof is done by combining these two inequalities. ■

A.3. Omitted Results for Sec. 5

Assumptions that Thm. 3 is relying on are following:

Assumption 2 (Regularity-1) Let $e \triangleq \{e_a(\cdot) : a \in \mathcal{A}\}$ be the true propensity score, $\vartheta \triangleq (\beta, \gamma)$ and $\vartheta_0 \triangleq (\beta_0, \gamma_0) \in \arg\min_{\beta,\gamma} \mathcal{R}^{\mathrm{db}}(\beta, \gamma; e)$.

1. **Positivity:** $e_a(x) \in [c, 1-c]$ for some constant $0 < c < 1/2$ for all $a, x \in \mathcal{A} \times \mathcal{X}$.
2. **f-divergence regularity:** f is convex and twice continuously differentiable; and the induced radius $B_f(e_a(x))$ is twice continuously differentiable on $[c, 1-c]$, with bounded first and second derivative; i.e., $\sup_{e\in[c,1-c]} |B_f(e)| + |B_f'(e)| + |B_f''(e)| < \infty$.
3. **Loss regularity:** For each fixed $e \in [c, 1-c]$, the map $\vartheta \mapsto \ell^{\mathrm{db}}(V; \vartheta, e)$ is twice continuously differentiable, with

$$\sup_{\vartheta,e} \|\ell^{\mathrm{db}}(V;\vartheta,e)\|_2^2 < \infty, \quad \sup_{\vartheta,e} \|\nabla_\theta \ell^{\mathrm{db}}(V;\vartheta,e)\|_2^2 < \infty, \quad \sup_{\theta,e} \|\nabla^2_{\vartheta\vartheta} \ell^{\mathrm{db}}(V;\vartheta,e)\|_2^2 < \infty.$$

4. **Higher-order smoothness:** Let $H(\vartheta; e) \triangleq \nabla^2_{\vartheta\vartheta} R^{\mathrm{db}}(\vartheta; e)$. There exists a neighborhood Θ_0 of ϑ containing ϑ_0 and constants $0 < \kappa \le \kappa_2 < \infty$ such that

$$\kappa_1 \mathbf{I} \;\preceq\; H(\vartheta; e) \;\preceq\; \kappa_2 \mathbf{I} \quad \text{for all } \vartheta \in \Theta_0. \tag{43}$$

5. **Uniform LLN:** For each fold k, define the empirical risk w.r.t. ℓ^{db} with the training fold is $\widehat{R}_k^{\mathrm{db}}(\vartheta, \widehat{e}^k)$. Then, we have a uniform law-of-large-number:

$$\sup_{\vartheta} \left|\widehat{R}_k^{\mathrm{db}}(\vartheta; \widehat{e}^k) - R^{\mathrm{db}}(\vartheta; \widehat{e}^k)\right| = O_p(n^{-1/2}), \tag{44}$$

Assumption 3 (Regularity-2) For $\vartheta \triangleq (\beta, \gamma)$, let $Z_\vartheta \triangleq g^*\left(\frac{\varphi(Y) - u_\gamma(A,X)\}}{\exp(h_\beta(A,X))}\right)$ and $m_\vartheta(a,x) \triangleq \mathbb{E}[Z_\vartheta \mid A = a, X = x]$. Let $\widehat{m}_k$ be the estimate for m_ϑ using the k-fold data.

1. **Bounded Nuisances:** $h_{\beta_0}, u_{\gamma_0}, h_{\hat{\beta}_k}, u_{\hat{\gamma}_k}$ are bounded by some constant M.
2. **Lipschitz parameterization:** The map $(\beta, \gamma) \mapsto (h_\beta(a,x), u_\gamma(a,x))$ is Lipschitz in $\vartheta = (\beta, \gamma)$ uniformly over all (a, x) with constant L_ϑ; i.e.,

$$\sup_{a,x} \left(|h_\beta(a,x) - h_{\beta'}(a,x)| + |u_\gamma(a,x) - u_{\gamma'}(a,x)|\right) \leq L_\vartheta \|\vartheta - \vartheta'\|. \tag{45}$$

3. **Smoothness of g^*:** The convex conjugate g^* is continuously differentiable with bounded derivative; i.e., $\sup_{t \in \mathcal{T}} |(g^*)'(t)| < \infty$ where $\mathcal{T}$ is a range where $g^*(t)$ is well-defined.
4. **Assumption on Regression:** $\|\widehat{m}_k - m_{\widehat{\vartheta}_k}\|_2 = O_p(s_n)$, where s_n is some sequence $s_n \to 0$; There exists a constant L_m s.t. $\|m_\vartheta - m_{\vartheta'}\|_2 \leq L_m \|\vartheta - \vartheta'\|$.
5. Correct Model Choice: Let $\overline{\theta}^\star_{\varphi,0}(a,x) \triangleq \mathbb{E}[\ell(V; (\beta_0, \gamma_0), e) \mid A = a, X = x]$. Then $\overline{\theta}^\star_{\varphi,0}(a,x) \triangleq \mathbb{E}[\ell(V; (\beta_0, \gamma_0), e) \mid A = a, X = x] = \overline{\theta}_\varphi(a,x)$ for all (a,x).

A.4. Specialization for $X = \emptyset$

In this section, we consider the case where $X = \emptyset$; i.e., we are interested in upper-limiting $Q_a(Y) \triangleq P(Y \mid \mathrm{do}(A = a))$ from the observational data $P_a(Y) \triangleq P(Y \mid A = a)$. We use $e_a \triangleq \Pr(A = a)$. Then, upper limits of statistical divergences between P_a and Q_a results are preserved, simply by putting $X = \emptyset$ as follow:

Corollary T1.4 (Specialization When $X = \emptyset$) Thm. 1, Coro. (T1.1, T1.2, T1.3), and Thm 2 hold with $X = \emptyset$, by replacing $P_{a,x}, Q_{a,x}, e_a(x)$ with P_a, Q_a, e_a, respectively.

Under the setting where $X = \emptyset$, the risk function is defined as follow:

Definition 6 (Risk Function for Causal Bounds When $X = \emptyset$) Let $h \triangleq \{h_a \in \mathbb{R}^+ : a \in \mathcal{A}\}$ and $u \triangleq \{u_a \in \mathbb{R}^+ : a \in \mathcal{A}\}$. Let $V \triangleq (A, Y)$ and $\eta_f^a \triangleq B_f(e_a)$. A risk function for

causal bound when $X = \emptyset$ is

$$\mathcal{R}(h, u; e) \triangleq \mathbb{E}_P[\ell(V; (h, u), \eta_f)], \tag{46}$$

where

$$\ell(V; (h, u), e) \triangleq \exp(h_A)\{\eta_f^a + g^\star(\tfrac{\varphi(Y) - u_A}{\exp(h_A)})\} + u_A. \tag{47}$$

Since estimating $e \triangleq \Pr(A = a)$ can be done in $o_P(n^{-1/2})$, we don't need the debiased risk if $X = \emptyset$. The causal bound estimator is defined as follow:

Definition 7 (Causal Bound Estimator when $X = \emptyset$) Fix a functional φ and an f-divergence. Let ℓ and $\mathcal{R}$ be as in Def. 6. Let the observed sample be i.i.d. $\{V_i \triangleq (A_i, Y_i)\}_{i=1}^n$. Define $n_a \triangleq \sum_{i=1}^n \mathbf{1}(A_i = a)$. The estimator of the upper causal bound $\overline{\theta}_\varphi(a)$ for any $a \in \mathcal{A}$ is constructed as follows:

1. Estimate the marginal propensity $\widehat{e}_a \triangleq n_a/n$.
2. Solve $\widehat{\vartheta} \triangleq (\widehat{h}, \widehat{u}) \in \arg\min_{h,u} \sum_{i=1}^n \ell(V_i; (h, u), \widehat{e})$.
3. Evaluate $\widehat{\lambda}_a \triangleq \exp(\widehat{h}_a)$.
4. Define the pseudo-outcome $\widehat{Z}_i \equiv g^*\Big(\frac{\phi(Y_i) - \widehat{u}_{A_i}}{\widehat{\lambda}_{A_i}}\Big)$ and evaluate $\widehat{m}_a \triangleq (1/n_a) \sum_{i:A_i=a} \widehat{Z}_i$.
5. Return $\widehat{\theta}_{\varphi,f}(a) \equiv \widehat{\lambda}_a\Big(\widehat{\eta}_{f,a} + \widehat{m}_a\Big) + \widehat{u}_a$, for $a \in \mathcal{A}$.

Before providing theoretical analysis, we provide a set of assumptions:

Assumption 4 (Regularity-3) Let $e \triangleq \{e_a : a \in \mathcal{A}\}$ where $e_a \triangleq \Pr(A = a)$, $\vartheta \triangleq (\beta, \gamma)$ and $\vartheta_0 \in \arg\min_\vartheta \mathcal{R}(\vartheta; e)$ where $\vartheta \triangleq (h, u) \triangleq \{(h_a, u_a) : a \in \mathcal{A}\}$. Let $Z_\vartheta \triangleq g^*(\frac{\varphi(Y) - u_A}{\exp(h_A)})$. Let $m_{\vartheta,a} \triangleq \mathbb{E}_{P_a}[Z_\vartheta]$.

1. **Positivity:** $e_a \in [c, 1 - c]$ for some constant $0 < c < 1/2$ for all $a \in \mathcal{A}$.
2. **f-divergence regularity:** f is convex and twice continuously differentiable; and the induced radius B_f is twice continuously differentiable on $[c, 1-c]$ with bounded derivatives; i.e., $\sup_{e \in [c,1-c]} |B_f(e)| + |B_f'(e)| + |B_f''(e)| < \infty$.
3. **Loss regularity:** For each fixed $e \in [c, 1 - c]$, the map $\vartheta \mapsto \ell(V; \vartheta, e)$ is twice continuously differentiable, with

$$\sup_{\vartheta,e} \|\ell(V; \vartheta, e)\|_2^2 < \infty, \quad \sup_{\vartheta,e} \|\nabla_\theta \ell(V; \vartheta, e)\|_2^2 < \infty, \quad \sup_{\theta,e} \|\nabla_{\vartheta\vartheta}^2 \ell(V; \vartheta, e)\|_2^2 < \infty.$$

4. **Higher-order smoothness:** Let $H(\vartheta; e) \triangleq \nabla^2_{\vartheta\vartheta} R(\vartheta; e)$. There exists a neighborhood Θ_0 of ϑ containing ϑ_0 and constants $0 < \kappa \leq \kappa_2 < \infty$ such that

$$\kappa_1 \mathbf{I} \preceq H(\vartheta; e) \preceq \kappa_2 \mathbf{I} \quad \text{for all } \vartheta \in \Theta_0. \tag{48}$$

5. **Uniform LLN:** Define the empirical risk w.r.t. ℓ with the training fold is $\widehat{R}(\vartheta, \widehat{e})$. Then, we have a uniform law-of-large-number:

$$\sup_{\vartheta} \left| \widehat{R}(\vartheta; \widehat{e}) - R(\vartheta; \widehat{e}) \right| = O_p(n^{-1/2}). \tag{49}$$

6. **Bounded Parameters:** h_a, u_a are bounded by some constant M.

7. **Smoothness of g^*:** The convex conjugate g^* is continuously differentiable with bounded derivative; i.e., $\sup_{t \in \mathcal{T}} |(g^*)'(t)| < \infty$ where $\mathcal{T}$ is a range where $g^*(t)$ is well-defined.

Theorem 4 (Error Analysis for Def. 7) Under the analogues of Assumption 4. Let $e \triangleq \{e_a : a \in \mathcal{A}\}$, and $\vartheta_0 \triangleq (h_0, u_0) \in \arg\min_{\vartheta} \mathcal{R}(\vartheta; e)$ for $\vartheta \triangleq (h, u)$. Let $\widehat{\vartheta} \triangleq (\widehat{h}, \widehat{u})$ be the minimizer in Step 2 in Def. 7 with $\widehat{e}$. Then,

$$\|\widehat{\vartheta} - \vartheta_0\|_2^2 = O_P(n^{-1/2}), \qquad \|\widehat{\theta}_\varphi - \overline{\theta}_\varphi\|_2^2 = O_P(n^{-1/2}). \tag{50}$$

Proof [**Proof of Thm. 4**] We first decompose the population excess risk as follow:

$$\begin{aligned}
R(\widehat{\vartheta}_k; e) - R(\vartheta_0; e) &= \underbrace{R(\widehat{\vartheta}; e) - R(\widehat{\vartheta}; \widehat{e})}_{(A)} && (51)\\
&= \underbrace{R(\widehat{\vartheta}; \widehat{e}) - \widehat{R}(\widehat{\vartheta}; \widehat{e})}_{(B)} && (52)\\
&= \underbrace{\widehat{R}(\widehat{\vartheta}\widehat{e}) - \widehat{R}(\vartheta_0; \widehat{e})}_{\leq 0} && (53)\\
&= \underbrace{\widehat{R}(\vartheta_0; \widehat{e}) - R(\vartheta_0; \widehat{e})}_{(C)} && (54)\\
&= \underbrace{R(\vartheta_0; \widehat{e}) - R(\vartheta_0; e)}_{(D)}. && (55)
\end{aligned}$$

It's obvious that $(B) + (C) = O_p(n^{-1/2})$ by the uniform LLN in Assumption 4.
To analyze $(A) + (D)$, fix a ϑ. Then,

$$\begin{aligned}
|R(\vartheta; \widehat{e}) - R(\vartheta; e)| &= \left| \mathbb{E}\left[\exp(h_A) \Big(B_f(\widehat{e}_A) - B_f(e_A) \Big) \right] \right| && (56)\\
&\leq \mathbb{E}\left[\exp(h_A) \left| B_f(\widehat{e}_A) - B_f(e_A) \right| \right]. && (57)
\end{aligned}$$

By boundedness parameters in Assumption 4, and B_f is Lipschitz on $[c, 1-c]$ with constant L_B with constant L_B,

$$|B_f(\hat{e}) - B_f(e)| \leq L_B |\widehat{e} - e|. \tag{58}$$

Thus, for any fixed ϑ,

$$\begin{aligned} |R(\vartheta; \widehat{e}) - R(\vartheta; e)| &\leq e^M L_B \sum_{a \in \mathcal{A}} e_a |\hat{e}_a - e_a| \leq e^M L_B \|\hat{e} - e\|_1 && (59) \\ &\leq e^M L_B \sqrt{2} \|\hat{e} - e\|_2 && (60) \\ &= O_P(n^{-1/2}), && (61) \end{aligned}$$

where $\|\hat{e} - e\|_2 = O_P(n^{-1/2})$ holds, because of the standard law of large number. Therefore, $(A) + (E) = O_P(n^{-1/2})$.
By Lemma 6,

$$\frac{\kappa_1}{2} \|\widehat{\vartheta} - \vartheta_0\|_2^2 \leq R(\widehat{\vartheta}; e) - R(\vartheta_0; e) = O_{\mathbb{P}}(n^{-1/2}). \tag{62}$$

Hence,

$$\|\widehat{\vartheta} - \vartheta_0\|_2^2 = O_{\mathbb{P}}(n^{-1/2}). \tag{63}$$

Continuing, we will denote

$$\begin{aligned} &\lambda_0 \triangleq \exp(h_a), \quad \lambda \triangleq \exp(\widehat{h}_a), \quad u \triangleq \widehat{u}_a && (64) \\ &\eta_0 \triangleq \eta_{f,a}, \quad \eta \triangleq \widehat{\eta}_{f,a} && (65) \\ &m_0 \triangleq \mathbb{E}_{P_a}[Z_i], \quad m \triangleq \widehat{m}_a. && (66) \end{aligned}$$

Continuing

$$\begin{aligned} \widehat{\theta}_{\varphi,f}(a) - \overline{\theta}_{\varphi,f}(a) &= \underbrace{(\lambda - \lambda_0)(\eta_0 + m_0)}_{(I)} && (67) \\ &+ \underbrace{(\lambda - \lambda_0)(\eta - \eta_0)}_{(II)} && (68) \\ &+ \underbrace{\lambda_0(\eta - \eta_0)}_{(III)} && (69) \\ &+ \underbrace{\lambda(m - m_0)}_{(IV)} && (70) \\ &+ \underbrace{(u - u_0)}_{(V)}. && (71) \end{aligned}$$

We note that

$$(I) + (V) \leq O_P(\|\widehat{\vartheta} - \vartheta_0\|_2^2) \leq O_{\mathbb{P}}(n^{-1/2}). \tag{72}$$

Consider (III).

$$|\text{(III)}| = |\lambda_0|\,|\eta - \eta_0| \leq e^M |\eta - \eta_0|, \tag{73}$$

where $|\eta - \eta_0| = |B_f(\widehat{e}_a) - B_f| \leq L_B \sum_{a\in\mathcal{A}} |\widehat{e}_a - e_a| = O_P(n^{-1/2})$.
Consider (II).

$$\|\text{(II)}\|_2^2 = O_P(\|\lambda_k - \lambda_0\|_2^2 \|\eta_k - \eta_0\|_2^2) = O_P(n^{-2}), \tag{74}$$

which implies that $\|\text{(II)}\|_2 = O_P(n^{-1})$.
Finally, consider (IV). Define

$$m_{\widehat{\vartheta}} \triangleq \mathbb{E}_{P_a}[\widehat{Z}_i]. \tag{75}$$

Then,

$$m - m_0 = (m - m_{\widehat{\vartheta}}) + (m_{\widehat{\vartheta}} - m_0). \tag{76}$$

By the law of large number, $(m - m_{\widehat{\vartheta}}) = O_P(n^{-1/2})$. Also, we have $(m_{\widehat{\vartheta}} - m_0) \leq L_m \|\widehat{\vartheta} - \vartheta_0\|$. Hence,

$$m_k - m_0 = O_P(n^{-1/2}) + O_P(\|\widehat{\vartheta}_k - \vartheta_0\|) = O_P(n^{-1/2}). \tag{77}$$

Therefore,

$$\|(IV)\|_2^2 = O_P(n^{-1/2}). \tag{78}$$

Combining, it shows that

$$\|\theta_k - \overline{\theta}_\varphi\|_2^2 = O_P(n^{-1/2}). \tag{79}$$

■

Appendix B. Omitted Preliminaries

B.1. Integral Probability Metrics and Related Quantities

This section provides formal definitions of discrepancy measures beyond the f-divergence family that are referenced in the main text, namely the mean difference, integral probability metrics (IPMs), and the maximum mean discrepancy (MMD).

Definition 8 (Mean Difference) Let P and Q be probability measures on a measurable space $(\mathcal{Y}, \mathcal{F})$ such that the first moments exist. The mean difference between P and Q is

defined as

$$D_{\mathrm{MD}}(P\|Q) \triangleq |\mathbb{E}_P[Y] - \mathbb{E}_Q[Y]| \,. \tag{80}$$

Definition 9 (Integral Probability Metric (IPM) (Müller, 1997)) Let P and Q be probability measures on a measurable space $(\mathcal{Y}, \mathcal{F})$, and let $\mathcal{H}$ be a class of measurable real-valued functions on $\mathcal{Y}$. The integral probability metric (IPM) between P and Q with respect to $\mathcal{H}$ is defined as

$$D_{\mathrm{IPM},\mathcal{H}}(P\|Q) \triangleq \sup_{h\in\mathcal{H}} |\mathbb{E}_P[h(Y)] - \mathbb{E}_Q[h(Y)]| \,. \tag{81}$$

The choice of the function class $\mathcal{H}$ determines the strength and properties of the resulting discrepancy measure. For instance, when $\mathcal{H}$ is restricted to linear functions of the form $h(y) = y$, the IPM reduces to the mean difference defined in Def. 8.

Definition 10 (Maximum Mean Discrepancy (MMD) (Gretton et al., 2012)) Let $\mathcal{H}_k$ be a reproducing kernel Hilbert space (RKHS) associated with a positive-definite kernel $k : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}$. The maximum mean discrepancy (MMD) between P and Q is defined as

$$D_{\mathrm{MMD},k}(P\|Q) \triangleq \sup_{\|h\|_{\mathcal{H}_k}\le 1} |\mathbb{E}_P[h(Y)] - \mathbb{E}_Q[h(Y)]| \,. \tag{82}$$

When the kernel k is characteristic, the MMD metrizes weak convergence, and $D_{\mathrm{MMD},k}(P\|Q) = 0$ if and only if $P = Q$. These quantities can be viewed as special cases of distributional discrepancies defined through restricted function classes, and serve as limiting or simplified alternatives to the f-divergence–based bounds considered in the main text.

Appendix C. Proofs

Proof of Thm. 1

We first declare some useful results:

Lemma 3 (f-divergence with Conditional Measure) Let P on $(\Omega, \mathcal{F})$ be an arbitrary probability measure. Let $E \in \mathcal{F}$ be a fixed event such that $P(E) = p \in (0, 1)$. Let $P_E(\cdot) \triangleq P(\cdot \mid E)$. Then,

$$D_f(P_E\|P) = pf(\tfrac{1}{p}) + (1-p)f(0) \tag{83}$$

Proof The Radon–Nikodým derivative of P_E w.r.t. P is

$$\frac{dP_E}{dP}(\omega) = \frac{1}{p}\mathbf{1}_E(\omega), \quad \forall \omega \in \Omega. \tag{84}$$

Recall that the Radon–Nikodým derivative $\frac{dP_E}{dP}(\omega)$ is defined as a function $g(\omega)$ satisfying the follows:

$$P_E(B) = \int_B g(\omega)P(d\omega), \quad \forall B \subset \mathcal{F}. \tag{85}$$

$g(\omega)$ is indeed $\frac{1}{p}\,1_E(\omega)$ because, for any fixed $B \subset \mathcal{F}$,

$$P_E(B) = \frac{1}{p}P_E(B)p = \frac{1}{p}P(B \cap E) = \int_B \frac{1}{p}\mathbf{1}_E(\omega)P(d\omega) = \int_B g(\omega)P(d\omega). \tag{86}$$

Then,

$$\begin{aligned} D_f(P_E\|P) &\triangleq \int f\Big(\frac{dP_E}{dP}(\omega)\Big)P(\omega) = \int f\Big(\frac{1}{p}\mathbf{1}(\omega)\Big)P(\omega) && (87)\\ &= \int_E f\Big(\frac{1}{p}\mathbf{1}(\omega)\Big)P(\omega) + \int_{E^c} f\Big(\frac{1}{p}\mathbf{1}(\omega)\Big)P(\omega) && (88)\\ &= pf(\tfrac{1}{p}) + (1-p)f(0). && (89) \end{aligned}$$

■

Lemma 4 (Data Processing Inequality (Csiszár, 1967)) Let P_X, Q_X denote a probability measure on $(\mathcal{X}, \mathcal{F}_X)$. Let $P_{Y|X}$ be a Markov kernel from $(\mathcal{X}, \mathcal{F}_X)$ to $(\mathcal{Y}, \mathcal{F}_Y)$. Let P_Y, Q_Y be the transformation of P_X, Q_X, respectively, when pushed through $P_{Y|X}$; i.e., $P_Y(B) = \int_{\mathcal{X}} P_{Y|X}(B \mid x)dP_X(x)$, and Q_Y is defined similarly. Then, for any f-divergence, we have

$$D_f(P_Y\|Q_Y) \leq D_f(P_X\|Q_X). \tag{90}$$

For any fixed $X = x$, set an event $E \triangleq \{A = a\}$ (i.e., A is fixed to a constant a). For a probability measure $P(U, A \mid x)$, we consider

$$P_E(U, A \mid x) \triangleq P(U, A \mid x, E) = P(U, A \mid x, a) = P(U \mid x, a). \tag{91}$$

Also, we note that

$$P(E \mid x) = P(a \mid x) = e_a(x). \tag{92}$$

Therefore, by Lemma 3,

$$
\begin{aligned}
& D_f(P_E(U, A \mid x) \| P(U, A \mid x)) && (93)\\
&= D_f(P(U \mid x, a) \| P(U, A \mid x)) && (94)\\
&= \underbrace{e_a(x) f(\tfrac{1}{e_a(x)}) + (1 - e_a(x)) f(0)}_{B(e_a(x), f)}. && (95)
\end{aligned}
$$

Define a transition kernel as follow: for a fixed $a \in \mathcal{A}$ and $x \in \mathcal{X}$,

$$P_{Y|U,A}(Y \mid U, A) \triangleq P(Y \mid U, a, x). \tag{96}$$

Then, we transformed $P(U, A \mid x)$ through the transitional kernel as follow:

$$\int P_{Y|U,A}(Y \mid U, A) P(u, a' \mid x) du da' = \int P(Y \mid u, a, x) P(u, a' \mid x) du da' = P(Y \mid \mathrm{do}(a), x). \tag{97}$$

Next, we transformed $P_E(U, A \mid x) = P(U \mid a, x)$ through the transitional kernel as follow:

$$\int P_{Y|U,A}(Y \mid U, A) P(u \mid a, x) du' = \int P(Y \mid u, a, x) P(u \mid a, x) du = P(Y \mid a, x). \tag{98}$$

By Lemma 4,

$$D_f(P(Y \mid a, x) \| P(Y \mid \mathrm{do}(a), x)) \leq D_f(P(U \mid a, x) \| P(U, A \mid x)) = B(e_a(x), f). \tag{99}$$

■.

Proof of Cor. T1.1

KL. With $f(t) = t \log t$ with $f(0) = 0$, we have

$$B(e_a(x), f) = -e_a(x) \frac{1}{e_a(x)} \log e_a(x) = -\log e_a(x). \tag{100}$$

Therefore,

$$D_{\mathrm{KL}}(P(Y \mid a, x) \| Q(Y \mid a, x)) \leq -\log e_a(x). \tag{101}$$

Hellinger. With $f(t) = \frac{1}{2}(\sqrt{t} - 1)^2$ with $f(0) = 1/2$, we have

$$
\begin{aligned}
B(e_a(x), f) &= e_a(x) f(\tfrac{1}{e_a(x)}) + (1 - e_a(x)) f(0) && (102)\\
&= \frac{1}{2} e_a(x) \Big(\sqrt{\tfrac{1}{e_a(x)}} - 1\Big)^2 + \frac{1}{2}(1 - e_a(x)) && (103)\\
&= 1 - \sqrt{e_a(x)}. && (104)
\end{aligned}
$$

To tighten, we use the following lemma:

Lemma 5 (Hellinger divergence vs. KL divergence) For any P, Q such that $P \ll Q$,

$$D_{\mathrm{H}}(P\|Q) \leq \frac{1}{2} D_{\mathrm{KL}}(P\|Q). \tag{105}$$

Proof We start with

$$D_{\mathrm{H}}(P\|Q) \triangleq \frac{1}{2}\int (\sqrt{p}(x) - \sqrt{q}(x))^2 dx = 1 - \int \sqrt{p(x)q(x)}dx. \tag{106}$$

Define $\mathtt{BC}(P,Q) \triangleq \int \sqrt{p(x)q(x)}dx$. Then, $D_{\mathrm{H}}(P\|Q) = 1 - \mathtt{BC}(P,Q)$. Define $D_{\mathrm{B}}(P\|Q) \triangleq -\log \mathtt{BC}(P,Q)$, which is known as Bhattacharyya distance.
Define $r(X) \triangleq \frac{q(x)}{p(x)}$. Then,

$$\mathtt{BC}(P,Q) = \int \sqrt{p(x)q(x)}dx = \int \sqrt{\tfrac{q(x)}{p(x)}}p(x)dx = \mathbb{E}_P\Big[\sqrt{r(X)}\Big]. \tag{107}$$

By Jensen's inequality, we have

$$\log \mathtt{BC}(P,Q) = \log \mathbb{E}_P\Big[\sqrt{r(X)}\Big] \geq \mathbb{E}_P\Big[\log \sqrt{r(X)}\Big] = \frac{1}{2}\mathbb{E}_P[\log r(X)]. \tag{108}$$

Also,

$$\mathbb{E}_P[\log r(X)] = \int p(x) \log \frac{q(x)}{p(x)} dx = -D_{\mathrm{KL}}(P,Q). \tag{109}$$

Combining,

$$-\frac{1}{2}D_{\mathrm{KL}}(P\|Q) \leq \log \mathtt{BC}(P,Q) \quad \Leftrightarrow \quad 1 - \exp\Big(-\frac{1}{2}D_{\mathrm{KL}}(P\|Q)\Big) \geq 1 - \mathtt{BC}(P,Q). \tag{110}$$

Finally,

$$D_{\mathrm{H}}(P\|Q) = 1 - \mathtt{BC}(P,Q) \leq 1 - \exp\Big(-\frac{1}{2}D_{\mathrm{KL}}(P\|Q)\Big) \leq \frac{1}{2}D_{\mathrm{KL}}(P\|Q), \tag{111}$$

where the last inequality holds since $1 - e^{-u} \leq u$ for any $u \geq 0$. ■

As a result, we can derive

$$D_{\mathrm{H}}(P(Y \mid a,x)\|Q(Y \mid a,x)) \leq -\frac{1}{2}\log e_a(x). \tag{112}$$

χ^2-divergence. Set $f(t) \triangleq \frac{1}{2}(t-1)^2$. Then, $B(e_a, f) = \frac{1-e_a(x)}{2e_a(x)}$.

Total variation. First, by Pinsker's inequality and the above inequality,

$$D_{\mathrm{TV}}(P\|Q) \leq \frac{1}{2}\sqrt{D_{\mathrm{KL}}(P\|Q)} \leq \frac{1}{2}\sqrt{-\log e_a(x)}. \tag{113}$$

By Bretagnolle–Huber bound (Bretagnolle and Huber, 1979) and the above inequality,

$$D_{\mathrm{TV}}(P\|Q) \leq \sqrt{1 - e^{D_{\mathrm{KL}}(P\|Q)}} \leq 1 - e_a(x). \tag{114}$$

■

Proof of Thm. 2

We note that

$$\mathbb{E}_{Q_{a,x}}[\varphi(Y)] = \mathbb{E}_{P_{a,x}}[s(Y)\varphi(Y)]. \tag{115}$$

Here, $P_{a,x} \ll Q_{a,x}$ guarantees $s(Y) > 0$. Since $P_{a,x}$ and $Q_{a,x}$ are distributions such that $P_{a,x} \ll Q_{a,x}$, $\mathbb{E}_{P_{a,x}}[s(Y)] = \int \frac{dQ_{a,x}}{dP_{a,x}} dP_{a,x} = 1$. Also,

$$\mathbb{E}_{P_{a,x}}[(g \circ s)] = \mathbb{E}_{P_{a,x}}[sf(1/s)] = \mathbb{E}_{Q_{a,x}}\left[\frac{dP_{a,x}}{dQ_{a,x}}\right] = D_f(P_{a,x}\|Q_{a,x}) \leq \eta_f(a,x). \tag{116}$$

Define the following Lagrangian function

$$\mathcal{L}(s,\lambda,u) = \mathbb{E}_P[s(Y)\varphi(Y)] + u(1 - \mathbb{E}_P[s(Y)]) + \lambda(\eta_f(a,x) - \mathbb{E}_P[g(s(Y))]) \tag{117}$$

$$= u + \lambda\eta_f(a,x) + \mathbb{E}_P\Big[s(Y)(\varphi(Y) - u) - \lambda g(s(Y))\Big]. \tag{118}$$

Standard Lagrange duality rewrites the upper bound as

$$\overline{\theta}_\varphi(a,x) = \inf_{\lambda \geq 0, u \in \mathbb{R}} \sup_{s>0} \mathcal{L}(s,\lambda,u), \tag{119}$$

$$= \inf_{\lambda \geq 0, u \in \mathbb{R}} \lambda\eta_f(a,x) + u + \sup_{s>0} \mathbb{E}_P\Big[s(Y)(\varphi(Y) - u) - \lambda g(s(Y))\Big], \tag{120}$$

under a suitable regularity condition (e.g., Slater condition). Next,

$$\sup_{s>0} \mathbb{E}_P\Big[s(Y)(\varphi(Y) - u) - \lambda g(s(Y))\Big] \tag{121}$$

$$= \sup_{s>0} \lambda\mathbb{E}_P\Big[s(Y)\tfrac{\varphi(Y)-u}{\lambda} - g(s(Y))\Big] \tag{122}$$

$$= \lambda \sup_{s>0} \mathbb{E}_P\Big[s(Y)\tfrac{\varphi(Y)-u}{\lambda} - g(s(Y))\Big] \tag{123}$$

$$= \lambda\mathbb{E}_P[g^*(\tfrac{\varphi(Y)-u}{\lambda})]. \tag{124}$$

Therefore,

$$\overline{\theta}_\varphi(a,x) = \inf_{\lambda \geq 0, u \in \mathbb{R}} \lambda\eta_f(a,x) + u + \lambda\mathbb{E}_P[g^*(\tfrac{\varphi(Y)-u}{\lambda})]. \tag{125}$$

■

Proof of Prop. 1

PROOF OF EQ. (15)

Define $r = 1/s$. Then,

$$g^*(t) \triangleq \sup_{s>0}\{st - g(s)\} = \sup_{r>0}\Big\{\frac{t}{r} - \frac{f(r)}{r}\Big\}. \tag{126}$$

Since f is a convex function such that $f \in \mathcal{C}^1((0,\infty))$, the supremum is achieved at r s.t.

$$\frac{d}{dr}\Big(\frac{1}{r}(t - f(r))\Big) = 0 \tag{127}$$

$$\Leftrightarrow -\frac{1}{r}t + \frac{1}{r}f(r) - f'(r) = 0 \tag{128}$$

$$\Leftrightarrow t = f(r) - rf'(r). \tag{129}$$

Define such map as $t(r) \triangleq f(r) - rf'(r)$. We note that this function has an inverse function since it's convex. Define the inverse function as $r(t)$. Note that the supremum occurred at $r = r(t)$. Therefore,

$$g^*(t) \triangleq \sup_{r>0}\Big\{\frac{t}{r} - \frac{f(r)}{r}\Big\} = \frac{1}{r(t)}(t - f(r(t))) = -\frac{r(t)}{r(t)}f'(r(t)) = -f'(r(t)), \tag{130}$$

where the last equation holds from $t = f(r) - rf'(r)$. ■

PROOF OF EQ. (16)

With $r \triangleq 1/s$, define $H_t(r) \triangleq \frac{t-f(r)}{r}$. We note that

$$g^*(t) \triangleq \sup_{s>0}\{st - sf(1/s)\} = \sup_{r>0}\Big\{\frac{t - f(r)}{r}\Big\} = \sup_{r>0} H_t(r). \tag{131}$$

Recall that

$$\partial f(r) \triangleq \{a:\ f(u) \geq f(r) + a(u - r)\ \ \forall u \geq 0\}. \tag{132}$$

For any fixed $r > 0$ and $a \in \partial f(r)$, and for any $v > 0$ and $u \triangleq 1/v$, the following holds:

$$vf(1/v) \geq f(r) + a(1 - rv) \quad \Rightarrow \quad \frac{f(r) - vf(1/v)}{1 - rv} \leq -a. \tag{133}$$

For studying the trajectory, we first approximate interior maximizer by finding the zero-subgradient.

$$\partial H_t(r) = \Big\{-\frac{t - f(r)}{r^2} - \frac{\partial f(r)}{r}\Big\} = \Big\{\frac{-ar + f(r) - t}{r^2}\Big\}. \tag{134}$$

Then, $\partial H_t(r) = 0$ means $t = f(r) - ra$. Therefore, we will check trajectories $\mathcal{T} \triangleq \{t = f(r) - ra,\ \forall r > 0,\ a \in \partial f(r)\}$. When $t \in \mathcal{T}$,

$$st - g(s) = st - sf(1/s) = \frac{t - f(r)}{r} = \frac{f(r) - ra - f(r)}{r} = -a. \tag{135}$$

When $t \notin \mathcal{T}$, then the supremum is realized by the endpoints; i.e., for $t \notin \mathcal{T}$

$$g^*(t) = \max\left\{ \limsup_{r \downarrow 0} \frac{t - f(r)}{r}, \ \limsup_{r \uparrow \infty} \frac{t - f(r)}{r} \right\}. \tag{136}$$

■

C.1. Proof of Prop. 3

((1) $\implies$ (2)). For each fixed (a, x),

$$\Delta(a, x) \triangleq \ell(h^\star, u^\star; a, x) - \inf_{h,u} \ell(h, u; a, x). \tag{137}$$

Assume, for contradiction, that (2) fails; i.e., $P_{A,X}(B) > 0$ for $B \triangleq \{(a, x) : \Delta(a, x) > 0\}$. For each $(a, x) \in B$, by the definition of the infimum, there exists a pair $\tilde{h}, \tilde{u}$ and $\epsilon(a, x) > 0$ such that

$$\ell(h^\star, u^\star; a, x) - \ell(\tilde{h}, \tilde{u}; a, x) > \epsilon(a, x) > 0. \tag{138}$$

Define $h'(a, x) \triangleq \tilde{h}(a, x)\mathbf{1}((a, x) \in B) + h^\star(a, x)\mathbf{1}((a, x) \notin B)$ and define $u'(a, x)$ similarly. Then,

$$\begin{aligned}
\mathcal{R}(h^\star, u^\star) &= \mathbb{E}[\ell(h^\star, u^\star, A, X)\mathbf{1}((A, X) \notin B)] + \mathbb{E}[\ell(h^\star, u^\star, A, X)\mathbf{1}((A, X) \in B)] && (139)\\
&> \mathbb{E}[\ell(h^\star, u^\star, A, X)\mathbf{1}((A, X) \notin B)] + \mathbb{E}[\ell(\tilde{h}, \tilde{u}, A, X)\mathbf{1}((A, X) \notin B)] && (140)\\
&= \mathcal{R}(h', u'). && (141)
\end{aligned}$$

This contradicts with the assumption (1). Therefore, $P_{A,X}(B) = 0$; i.e., $(h^\star, u^\star)$ is a minimizer of $\ell(h, u; a, x)$ for $P_{A,X}$-almost every (a, x).

((2) $\implies$ (1)). Since $\ell(h^\star, u^\star; a, x) < \ell(h, u; a, x)$ for all (h, u) and for $P_{A,X}$-almost every a, x, $\mathcal{R}(h^\star, u^\star) < \mathcal{R}(h, u)$.

Proof of Lemma 1

Define

$$\begin{aligned}
e_a &\triangleq \Pr(A = a \mid X) && (142)\\
\lambda_a &\triangleq \exp(h_\beta(a, X)) && (143)\\
B_a(e) &\triangleq B_f(e_a(X)) && (144)\\
u_a &\triangleq u(a, X) && (145)\\
g_a^* &\triangleq g^*\left(\frac{\varphi(Y) - u(A, X)}{\lambda_A(X)}\right). && (146)
\end{aligned}$$

Then,

$$\ell(V; (\beta, \gamma), e) \triangleq \lambda_A(B_A + g_A^*) + u_A(X). \tag{147}$$

Define

$$L_1(e) \triangleq \lambda_A(B_A + g_A^*) + u_A(X). \tag{148}$$

The correction term is

$$L_2(e) \triangleq \sum_a e_a \lambda_a B_a' \{\mathbf{1}(A = a) - e_a\}. \tag{149}$$

Then, $\mathcal{R}^{\mathrm{db}}(e) \triangleq \mathbb{E}[L_1(e) + L_2(e)]$. Then,

$$\frac{\partial}{\partial t}\mathbb{E}[L_1(e_t)]\Big|_{t=0} = \frac{\partial}{\partial t}\mathbb{E}[L_1(e_A + ts_A)]\Big|_{t=0} \tag{150}$$

$$= \mathbb{E}\left[\lambda_A B'(e_A) s_A\right]. \tag{151}$$

Also,

$$L_2(e) \triangleq \sum_a \underbrace{e_a \lambda_a B_a'}_{U_a(e_a)} \underbrace{\{\mathbf{1}(A = a) - e_a\}}_{V_a(e_a)}. \tag{152}$$

Then,

$$\frac{\partial}{\partial t}\mathbb{E}[L_1(e_t)]\Big|_{t=0} = \mathbb{E}\left[\sum_{a\in\mathcal{A}}\left(\frac{\partial U_a}{\partial t}V_a + \frac{\partial V_a}{\partial t}U_a\right)\right], \tag{153}$$

where

$$\mathbb{E}\left[\sum_{a\in\mathcal{A}} U_a'(e)V_a(e_a)\right] = \mathbb{E}_X\left[\sum_{a\in\mathcal{A}} U_a'(e)\mathbb{E}_{A|X}[\mathbf{1}(A = a) - e_a]\right] = 0, \tag{154}$$

and

$$\mathbb{E}\left[\sum_{a\in\mathcal{A}} U_a(e)V_a'(e_a)\right] = -\mathbb{E}\left[\sum_{a\in\mathcal{A}} U_a(e)s_a\right] = -\mathbb{E}_X\left[\sum_{a\in\mathcal{A}} e_a\lambda_a B_a' s_a\right]. \tag{155}$$

Then,

$$\frac{\partial R^{\mathrm{db}}}{\partial t} = \mathbb{E}[\lambda_A B'(e_A)s_A] - \mathbb{E}_X\left[\sum_{a\in\mathcal{A}} e_a\lambda_a B_a' s_a\right] \tag{156}$$

$$= \mathbb{E}_X\left[\sum_{a\in\mathcal{A}} e_a\lambda_a B_a' s_a\right] - \mathbb{E}_X\left[\sum_{a\in\mathcal{A}} e_a\lambda_a B_a' s_a\right] \tag{157}$$

$$= 0. \tag{158}$$

Proof of Theorem 3

Lemma 6 (Higher-order smoothness $\Leftrightarrow$ Local quadratic expansion inequality) Higher-order smoothness in Assumption 2 is equivalent to the local quadratic expansion inequality:

$$\frac{\kappa_1}{2}\|\vartheta-\vartheta_0\|^2 \;\le\; \mathcal{R}^{\mathrm{db}}(\vartheta;e_0)-\mathcal{R}^{\mathrm{db}}(\vartheta_0;e_0) \;\le\; \frac{\kappa_2}{2}\|\vartheta-\vartheta_0\|^2, \text{ for } \vartheta\in\Theta_0. \tag{159}$$

Proof [Proof of Lemma 6] ($\Rightarrow$). Let $r(t) \triangleq \mathcal{R}(\vartheta_t;e_0)$, where $\vartheta_t \triangleq \vartheta_0 + t(\vartheta-\vartheta_0)$ for $t\in[0,1]$. By the chain rule,

$$r'(t) \triangleq (\vartheta-\vartheta_0)^\intercal \nabla_\vartheta \mathcal{R}(\vartheta_t;e_0) \tag{160}$$

$$r''(t) \triangleq (\vartheta-\vartheta_0)^\intercal \nabla_{\vartheta\vartheta} H(\vartheta_t;e_0)(\vartheta-\vartheta_0). \tag{161}$$

Since $\nabla_\vartheta \mathcal{R}(\vartheta_t;e_0) = 0$ (since ϑ_0 is a local minimizer), by the fundamental theorem of calculus, we have

$$\mathcal{R}(\vartheta;e_0)-\mathcal{R}(\vartheta_0;e_0) = \int_0^1 (1-t)(\vartheta-\vartheta_0)^\intercal H(\vartheta_t;e_0)(\vartheta-\vartheta_0)dt. \tag{162}$$

Now, applying the Higher-order smoothness, we have

$$\frac{\kappa_1}{2}\|\vartheta-\vartheta_0\|_2^2 \le \int_0^1 (1-t)(\vartheta-\vartheta_0)^\intercal H(\vartheta_t;e_0)(\vartheta-\vartheta_0)dt \le \frac{\kappa_2}{2}\|\vartheta-\vartheta_0\|_2^2. \tag{163}$$

($\Leftarrow$). Assume Eq. (159). Consider more general $r(t)$ as $r(t) \triangleq \mathcal{R}(\vartheta_0+t\vartheta;e_0)$. By Eq. (159),

$$\frac{\kappa_1}{2}t^2\|v\|^2 \le r(t)-r(0) \le \frac{\kappa_2}{2}t^2\|v\|^2. \tag{164}$$

Since $r'(0)=0$, Taylor expansion gives

$$r(t)-r(0) = \frac{1}{2}r''(0)t^2 + o(t^2), \implies \kappa_1\|\vartheta\|^2 \le \vartheta^\intercal H(\vartheta_0;e_0)\vartheta \le \kappa_2\|\vartheta\|^2. \tag{165}$$

■

C.1.1. Proof of Eq. (26)

For brevity, we write $R(\vartheta;e') \triangleq R^{\mathrm{db}}(\vartheta;e')$ for any ϑ and e'. Let $\widehat{R}_k$ denote the empirical risk of R using the k'th fold dataset.

We first decompose the population excess risk as follow:

$$R(\widehat{\vartheta}_k; e_0) - R(\vartheta_0; e_0) = \underbrace{R(\widehat{\vartheta}_k; e_0) - R(\widehat{\vartheta}_k; \widehat{e}^k)}_{(A)} \tag{166}$$

$$= \underbrace{R(\widehat{\vartheta}_k; \widehat{e}^k) - \widehat{R}_k(\widehat{\vartheta}_k; \widehat{e}^k)}_{(B)} \tag{167}$$

$$= \underbrace{\widehat{R}_k(\widehat{\vartheta}_k; \widehat{e}^k) - \widehat{R}_k(\vartheta_0; \widehat{e}^k)}_{\leq 0} \tag{168}$$

$$= \underbrace{\widehat{R}_k(\vartheta_0; \widehat{e}^k) - R(\vartheta_0; \widehat{e}^k)}_{(C)} \tag{169}$$

$$= \underbrace{R(\vartheta_0; \widehat{e}^k) - R(\vartheta_0; e_0)}_{(D)}. \tag{170}$$

We will show that

1. $(B) + (C) = O_p(n^{-1/2})$ by the uniform LLM in Assumption 2.
2. $(A) + (D) = O_p(r_n^2)$ by the orthogonality and smoothness in Assumption 2.

As a result,

$$R(\widehat{\vartheta}_k; e_0) - R(\vartheta_0; e_0) = O_p(n^{-1/2}) + O_p(r_n^2). \tag{171}$$

Bounds for $(B) + (C)$. Terms $(B) + (D)$ are bounded by Unfiform LLN as follow:

$$(B) + (D) \leq 2 \sup_{\vartheta} \left| R(\vartheta; \widehat{e}^k) - \widehat{R}_k(\vartheta; \widehat{e}^k) \right| = O_p(n^{1-2}). \tag{172}$$

Bounds for $(A) + (D)$. Fix a ϑ. Consider a parametric submodel $t \mapsto e^t \triangleq e_0 + t(\widehat{e}^k - e_0)$. Let $\delta e_0 \triangleq \widehat{e}^k - e_0$.

Then, by Taylor's expansion and the mean-value theorem,

$$R(\vartheta; \widehat{e}^k) = R(\vartheta; e_0) + \nabla_e R(\vartheta; e_0)[\delta e_0] + \frac{1}{2} \nabla_{ee} R(\vartheta; e^\dagger)[\delta e_0, \delta e_0], \tag{173}$$

for some $e^\dagger$ on the line between e_0 and $\hat{e}^k$. This is equivalent to

$$R(\vartheta; e_0) - R(\vartheta; \widehat{e}^k) = -\nabla_e R(\vartheta; e_0)[\delta e_0] - \underbrace{\frac{1}{2} \nabla_{ee} R(\vartheta; e^\dagger)[\delta e_0, \delta e_0]}_{O_P(\|\widehat{e}^k - e_0\|_2^2)}, \tag{174}$$

where $\nabla_{ee} R(\vartheta; e^\dagger)[\delta e_0, \delta e_0] = O_P(\|\widehat{e}^k - e_0\|_2^2)$ holds by the boundedness of the second derivative $\nabla_{ee} R$ in the loss regularity in Assumption 2.

By Lemma 1, $\nabla_e R(\vartheta_0; e_0)[\delta e_0] = 0$. Therefore,

$$(D) = O_P(\|\widehat{e}^k - e_0\|_2^2) = O_P(r_n^2). \tag{175}$$

Now, consider Term $(A) = R(\widehat{\vartheta}_k; e_0) - R(\widehat{\vartheta}_k; \widehat{e}^k)$.

$$(A) = -\nabla_e R(\widehat{\vartheta}_k; e_0)[\widehat{e}^k - e_0] + O_P(\|\widehat{e}^k - e_0\|_2^2), \tag{176}$$

where

$$\nabla_e R(\widehat{\vartheta}_k; e_0)[\widehat{e}^k - e_0] = \nabla_e R(\vartheta_0; e_0)[\widehat{e}^k - e_0] + \nabla_e \nabla_\vartheta R(\vartheta_0, e_0)[\widehat{\vartheta}_k - \vartheta_0, \widehat{e}^k - e_0], \tag{177}$$

where $\nabla_e R(\vartheta_0; e_0)[\widehat{e}^k - e_0] = 0$ by Lemma 1 and

$$\nabla_e \nabla_\vartheta R(\vartheta_0, e_0)[\widehat{\vartheta}_k - \vartheta_0, \widehat{e}^k - e_0] = O_P(\|\widehat{\vartheta}_k - \vartheta_0\|_2 \|\widehat{e}^k - e_0\|_2) \tag{178}$$

by the boundedness of the derivative. As a result,

$$(A) = O_P(\|\widehat{\vartheta}_k - \vartheta_0\|_2 \cdot r_n) + O_P(r_n^2), \tag{179}$$

and

$$(A) + (D) = O_P(\|\widehat{\vartheta}_k - \vartheta_0\|_2 \cdot r_n) + O_P(r_n^2). \tag{180}$$

Bound Derivation. Combining,

$$R(\widehat{\vartheta}_k; e_0) - R(\vartheta_0; e_0) = O_p(n^{-1/2}) + O_P(\|\widehat{\vartheta}_k - \vartheta_0\|_2 \cdot r_n) + O_P(r_n^2). \tag{181}$$

From Lemma 6, we have

$$\frac{\kappa_1}{2}\|\widehat{\vartheta}_k - \vartheta_0\|^2 \leq R(\widehat{\vartheta}_k; e_0) - R(\vartheta_0; e_0) = O_p(n^{-1/2}) + O_P(\|\widehat{\vartheta}_k - \vartheta_0\|_2 \cdot r_n) + O_P(r_n^2). \tag{182}$$

This means that there exists two constant $C_1, C_2 < \infty$ s.t.

$$\frac{\kappa_1}{2}\|\widehat{\vartheta}_k - \vartheta_0\|^2 - C_1 r_n \|\widehat{\vartheta}_k - \vartheta_0\|_2 \leq C_2(n^{-1/2} + r_n^2). \tag{183}$$

By the simple algebra, we have

$$\|\widehat{\vartheta}_k - \vartheta_0\| \leq \frac{1}{\kappa_1}\Big\{C_1 r_n + \sqrt{(C_1 r_n)^2 + 2\kappa_1 C_2(n^{-1/2} + r_n^2)}\Big\}, \tag{184}$$

which shows that

$$\|\widehat{\vartheta}_k - \vartheta_0\|^2 = O_p(n^{-1/2} + r_n^2). \tag{185}$$

C.1.2. PROOF OF EQ. (27)

For brevity, we just write

$$\theta_k \triangleq \widehat{\theta}_{\varphi}^{(k)}, \quad \lambda_k \triangleq \widehat{\lambda}_k, \quad \eta_k \triangleq \widehat{\eta}_f^k, \quad m_k \triangleq \widehat{m}_k, \quad u_k \triangleq \widehat{u}_k. \tag{186}$$

All the true parameters are indexed as 0. For each (a, x),

$$\theta_k(a,x) - \overline{\theta}_{\varphi}(a,x) = \underbrace{(\lambda_k - \lambda_0)(\eta_0 + m_0)}_{(I)} \tag{187}$$

$$+ \underbrace{(\lambda_k - \lambda_0)(\eta_k - \eta_0)}_{(II)} \tag{188}$$

$$+ \underbrace{\lambda_0(\eta_k - \eta_0)}_{(III)} \tag{189}$$

$$+ \underbrace{\lambda_k(m_k - m_0)}_{(IV)} \tag{190}$$

$$+ \underbrace{(u_k - u_0)}_{(V)}. \tag{191}$$

We first bound (V). By Lipschitz parametrization in Assumption 3, we have

$$\|u_k - u_0\|_2^2 \leq L_{\vartheta}^2 \|\widehat{\vartheta}_k - \vartheta_0\|_2^2. \tag{192}$$

That is,

$$\|(V)\|_2^2 = O_P(\|\widehat{\vartheta}_k - \vartheta_0\|_2^2). \tag{193}$$

Now consider (I) part. By the bounded nuisance in Assumption 3, we have

$$\|\lambda_k - \lambda_0\|_2^2 \leq C_0 e^M L_{\vartheta} \|\widehat{\vartheta}_k - \vartheta_0\|_2^2, \tag{194}$$

for some constant C_0. Therefore,

$$\|(I)\|_2^2 = O_P(\|\widehat{\vartheta}_k - \vartheta_0\|_2^2). \tag{195}$$

Consider (III).

$$|(\text{III})(a,x)| = |\lambda_0(a,x)|\,|\eta_k(a,x) - \eta_0(a,x)| \leq e^M |\eta_k(a,x) - \eta_0(a,x)| = O_P(r_n^2), \tag{196}$$

under the smoothness assumption on B_f.

Consider (II).

$$\|(\text{II})\|_2^2 = O_P(\|\lambda_k - \lambda_0\|_2^2 \|\eta_k - \eta_0\|_2^2). \tag{197}$$

Finally, consider (IV). Define

$$m_{\widehat{\varphi}}(a, x) \triangleq \mathbb{E}[Z_i^k \mid A = a, X = x]. \tag{198}$$

Then,

$$m_k - m_0 = (m_k - m_{\widehat{\vartheta}}) + (m_{\widehat{\vartheta}} - m_0). \tag{199}$$

By Assumption 3, $(m_k - m_{\widehat{\vartheta}}) = O_P(s_n)$. By Lipschitz, we have $(m_{\widehat{\vartheta}} - m_0) \leq L_m \|\widehat{\vartheta} - \vartheta_0\|$. Hence,

$$m_k - m_0 = O_P(s_n) + O_P(\|\widehat{\vartheta}_k - \vartheta_0\|). \tag{200}$$

Therefore,

$$\|(IV)\|_2^2 \leq e^{2M}(O_P(s_n) + O_P(\|\widehat{\vartheta}_k - \vartheta_0\|))^2 = O_P(s_n^2) + O_P(\|\widehat{\vartheta}_k - \vartheta_0\|^2). \tag{201}$$

Combining, it shows that

$$\|\theta_k - \overline{\theta}_\varphi\|_2^2 = O_P(n^{-1/2} + r_n^2 + s_n^2). \tag{202}$$

C.2. Proof of Lemma 2

Suppose k upper bound estimates is correct. This means that $\widehat{\theta}_{\text{up},f_1}^{(n_f)} \geq \cdots \geq \widehat{\theta}_{\text{up},f_{n_f-k+1}}^{(n_f-k+1)} \geq \theta$. This implies that, the k'th largest upper bound estimates, which is $\widehat{\theta}_{\text{up},f_{n_f-k+1}}^{(n_f-k+1)}$, is correctly upper bounding the true effect.

Suppose k lower bound estimates is correct. This means that $\widehat{\theta}_{\text{lo},f_1}^{(1)} \leq \cdots \leq \widehat{\theta}_{\text{lo},f_k}^{(k)} \leq \theta$. This implies that, the k'th smallest lower bound estimates, which is $\widehat{\theta}_{\text{lo},f_k}^{(k)}$, is correctly lower bounding the true effect.